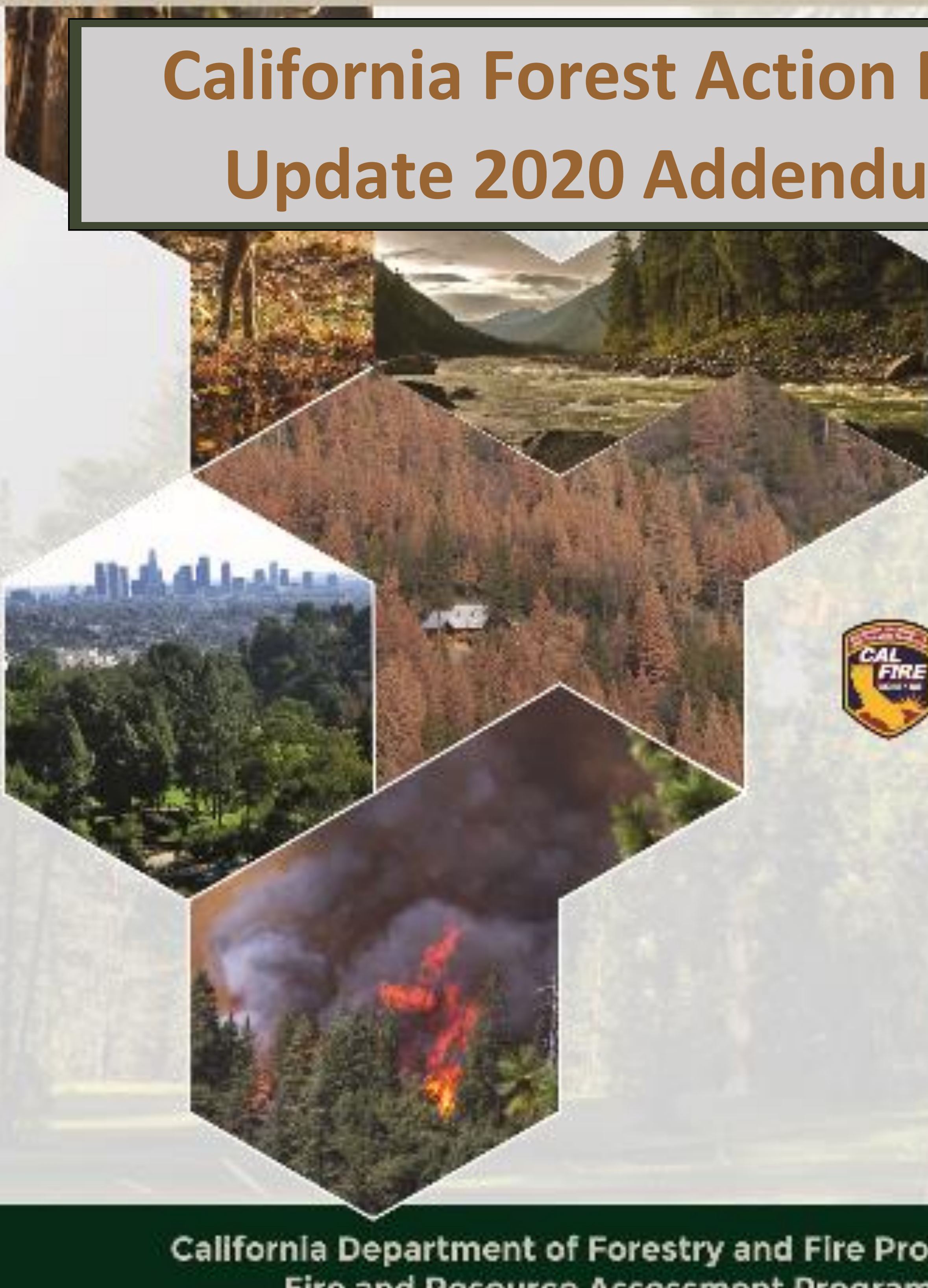
CALIFORNIA'S FORESTS AND RANGELANDS

2017 ASSESSMENT

California Forest Action Plan Update 2020 Addendum



California Department of Forestry and Fire Protection
Fire and Resource Assessment Program

California Forest Action Plan Update 2020 Addendum:

Connecting People with Trees, Forests, and Open Space

Key Findings

Section 1: CAL FIRE programs and challenges

- Collaborative forest or watershed management assessment and enhancement projects are increasingly scaling up to the landscape level, thus engaging more and more landowners, communities, agencies, and nongovernmental organizations. This is expanding people's connection with trees to the level of landscape-level ecosystem processes.
- CAL FIRE's Urban and Community Forestry Program positively touches the lives of millions of Californians every day. For example, in fiscal year 2019, over 33 million urbanites were living in communities that received educational, technical, and/or financial assistance from the Program.
- In recent years, most Californians have felt a direct connection to their forests and other wildlands in unfortunate ways, through direct or indirect experience of highly destructive fires, wildlife smoke, highly disruptive public safety power shutoffs, and other impacts. Community and landowner engagement in wildfire protection planning and projects, with backing from State and Federal agencies, is increasingly coalescing to address this situation, but much remains to be done in the face of a changing climate that is a driver for major fire events.

Section 2: Connecting People to Trees

- Given the increase of nature deficit disorder, providing opportunities to experience nature in larger natural parks, forests and rangelands is essential to public health, particularly to those in in urban areas.
- COVID-19, and the resulting economic hardships, have put the majority of outdoor education schools in California at risk of permanent closure.
- Outdoor education has become increasingly expensive, leading to inequitable access for lower income communities, particularly in largely underrepresented groups in outdoor education.
- Many outdoor education schools are located in high wildfire risk areas, putting their property at risk of destruction as well as necessitating the protection of properties from wildfires through defensible space and updating structures.
- Prior to COVID-19, major challenges facing outdoor education programs have been the transportation costs and liability issues of children traveling to distant outdoor schools.

Section 3: Trends in Publicly Accessible Forests and Open Space

- Overall, publicly accessible acreage in natural areas varies greatly in among counties and essentially is inversely proportional to the respective percentages of development in urban and agricultural uses. Unfortunately, access to natural landscapes for some urban populations is quite limited.
- Data from the California Protected Areas Database (CPAD) indicate that from 2012 to 2020, just over 532,000 acres of forest and rangeland have been added statewide to the land base of undeveloped natural and semi-natural areas publicly accessible as parks and open space.

- On average over the 8-year period, about 9,200 acres of parks and open space have been added per county, which is a rate of about 1,148 acres on average per county per year. The amount of increase has, however, varied greatly among counties. The range was from over 81,000 acres added in Kern County, to a 34-acre increase in Sutter County.

Introduction

FRAP has prepared this addendum to comply with the Forest Service requirement that State Forest Resource Assessments address the National Priority "Connecting People with Trees, Forests, and Open Space." This addendum begins by tying back to several sections in the 2017 FRAP Assessment that touch on this area. Second, it looks to quantify trends in public and (mainly) student participation in programs that enable extended contact with the outdoors, trees, and natural areas. Third, it examines trends in the availability of natural open space to people in California, on a county basis. Together these three areas provide indications of progress toward opening more opportunities for the engagement of the general populace to experience the great natural out-of-doors.

Section 1: Tying Back to Other Sections of California's Forests and Rangelands 2017 Assessment

At least four chapters in the 2017 Assessment address the topic of connecting people with trees, forests, and open space:

- Chapter 1, Sustainable Working Forests
- Chapter 3, Urban Forestry
- Chapter 4, Wildfire
- Chapter 11, Reducing Community Wildfire Risks

While not addressed in detail in the Assessment, CAL FIRE's nine Demonstration State Forests (totaling some 72,000 acres) provide an important opportunity for Californians to connect to trees, forests, and open space. Their programs for recreation (e.g., camping, hiking, horseback riding, mountain biking) and educational demonstration of forest management techniques all help with connecting people with forests. For example, Soquel Demonstration State Forest in Santa Cruz County is a regional magnet for mountain bikers. Mountain Home Demonstration State Forest, located in the Southern Sierra Nevada in Tulare County, draws people from the San Joaquin Valley and Southern California for camping and hiking. An important attraction in both forests is the presence of redwoods, with Coast redwoods on Soquel and giant sequoia (including old growth) on Mountain Home. The Demonstration State Forests are often used as teaching sites for educational programs targeted to small, private forestland owners.

Assessment Chapter 1, Sustainable Working Forests, addresses the topic of connecting people with trees and forests through its discussion of technical and financial assistance that is provided to small, private forestland owners. For example, engaging these owners in their forest through the preparation of a Forest Stewardship Plan is an excellent way to connect them to their forest in a fashion that incorporates their own, personal values for the management of these lands. Private forest landowners also can be engaged though working forest conservation easements, and through collaborative projects varying from the scale of a few landowners to the watershed or landscape scale. CAL FIRE also engages private forest landowners in forestry through the research and educational programs of its Demonstration State Forests.

Relevant Key Findings from Chapter 1 of the Assessment are:

- Conservation easements are an increasingly effective tool for preserving timberlands with important environmental or social values, and for protecting working forests from conversion or being subdivided.
- Several collaborative projects involving local communities, forest managers and environmental groups exist in forested regions. The most comprehensive of these is the Sierra Nevada Conservancy "Sierra Nevada Forest and Community Initiative" [now called the "Watershed Improvement Program"]. These projects are indicative of a general interest in working at the landscape scale to promote forest health and the sustainability of rural communities.
- Federal and State programs continue to be important sources of technical and financial assistance to small landowners for their forest management planning and activities.

Relevant Opportunities identified in Chapter 1 of the Assessment are:

- Continue to Explore, Implement, and Support Ways to Increase Active Sustainable Management on Nonindustrial Timberlands.
- Support Collaborative Landscape-Level Projects that Involve State and Federal Agencies, Local Communities, and other Stakeholders for Improving Economic and Environmental Sustainability.

Relevant Indicators from Chapter 1 and 6 of the Assessment are:

- 1.5. Timberland Managed Under Forest Certification or Other Sustainable Forestry Standards.
- 6.3. Private Forest and Rangeland Under Easements of Conservation Organization Owned.

Assessment Chapter 3, Urban Forestry, shows that urban trees, the urban forest, and urban and community forest programs (at the state, regional, or local level) can be powerful mechanism for connecting people to trees, forests, and open space. Since about 95% of California residents live in cities, urban forestry may be our best opportunity to foster this connection at the greatest scale. Fortunately, more communities are realizing that urban forests are a critical resource and that increased investment and planning to manage this resource are needed.

Relevant to the Federal goal of connecting people to trees, forests, and open space, the current CAL FIRE Urban and Community Forestry Program Strategic Plan includes objectives to:

- Increase awareness of the value of urban forests for all Californians.
- Make available urban forestry resources to local decision-makers and the public.
- Encourage local decision makers to fund urban forests as essential community infrastructure.

Environmental conditions and elements of urban area, including tree canopy cover, access to open space, and density of impervious surfaces impact the quality of life of the residents, and can vary greatly across regions, counties, and even local jurisdictions. Urban forestry efforts that target specific neighborhoods' needs can often provide the most benefits. CAL FIRE's programs place significant emphasis on serving under-resourced, low-income communities to increase the urban forest canopy and associated benefits. This preference is called for in the state's Urban Forestry Act.

Chapter 3 of the Assessment looks at the level of urban tree canopy as a particularly important factor for the evaluation of urban forests. Importantly, it finds that:

> The uneven distribution of UTC [urban tree canopy] across the state and within individual cities leads to an inequitable distribution of environmental benefits. Additional considerations for future analyses would include disadvantaged communities where the environmental benefits from urban trees are often most needed.

Over 61% of urban areas in the state have less than 10% tree canopy. American Forests has established an urban tree canopy goal of a minimum of 25%.

The section in Chapter 3 of the Assessment on urban forestry commitments by county looks to urban forestry grants, Tree City USA designations, and CARS reporting as metrics of urban forestry programs activities. These programs all help to connect urban residents to trees, forests, and open spaces. CAL FIRE's Urban and Community Forestry Program has multiple grant categories, all of which serve to connect residents to their urban forests in some ways. As one example of the scope of these activities, in fiscal year 2019-20, CAL FIRE awarded over $18.5 million in urban and community forestry grants, which by nature serve to connect residents to their urban forests. Many of these grants—made to cities, local urban forestry groups, or other nongovernmental organizations—were for projects that will take place in and will directly engage disadvantaged communities.

CAL FIRE's annual Urban and Community Forestry Program reporting to the Forest Service provides metrics for many of the program accomplishments. Particularly relevant for the theme of connecting people to trees and forests are the following from the 2019 fiscal year:

Number of people living in communities provided educational, technical and/or financial assistance.	33,416,462
Number of people living in communities that are developing programs/activities for their urban and community trees and forests.	20,169,260
Number of hours of volunteer service logged.	130,860

Relevant Key Findings from Chapter 3 of the Assessment are:

- There are approximately 9.1 million urban street trees, about 1 for every 4 residents.
- Urban tree canopy is not evenly distributed, and 61.4% of urban areas have less than 10% tree canopy cover.
- Average Urban Tree Canopy (UTC) varies from 3% in Imperial County (in the Sonoran Desert), to 66% in Tuolumne County.
- Of California's 58 Counties, 55 have urban areas, and only 12 of these exceed the 25% UTC American Forests goal.
- For all of California's 211 census-defined urban areas, the average statewide UTC cover is 15%. Only 44 of these exceed the American Forests goal of 25% UTC.

Relevant Indicators from Chapter 3 of the Assessment are:

- 3.1 Tree Canopy Cover

Assessment Chapter 4, Wildfire, and Assessment Chapter 11, Reducing Community Wildfire Risks, are directly connected and thus discussed here in an integrated fashion. Chapter 4 addresses the condition of the fire-prone landscapes of California, how wildfires are acting on natural systems in recent decades, and what strategies will be required to meet the fire management challenges of the 21st century. Chapter 11 provides a synthesis of indicators, issues, and opportunities for reducing wildfire risk to communities.

Relevant Key Findings from Chapters 4 and 11 of the Assessment are:

- More than 25 million acres in California (32% of burnable wildland vegetation) is classified as Very High or Extreme Fire Threat.
- Annual rates of burning in forest and shrub-dominated vegetation and average size of large fires (>1000 acres) have increased significantly over the last 17 years.
- In 2010, in all counties, about 3 million housing units (HU) were in Fire Hazard Severity Zones (FHSZ) and potentially at risk from wildfire. This includes about 1.2 million HU (41%) in the Very High class.
- Over 460,000 HUs were added within FHSZ between 2000 and 2010. This includes 144,000 HU added to the Very High class.
- There are 1,338 individual communities represented by the Communities at Risk (CAR) list. Of these communities, 66% (881) are covered by a CWPP (individual, regional or countywide) and/or are recognized by the Firewise program. Numerous other communities are at various stages of CWPP development.
- Of the CARs communities, 16% (213) are covered by individual CWPPs or the Firewise program. Individual CWPPs typically provide the finest detail for project-level planning; however, many county-level plans are very detailed, while others serve more generally as an umbrella for individual CWPPs.

Relevant Indicators from Chapters 4 and 11 of the Assessment are:

- 4.2 Fire Threat
- 4.3 Wildfire Activity—Trends in Area Burned
- 4.5 Fuel Treatment Area
- 11.2 Housing Units by Fire Hazard Severity Zone Class
- 11.3 Housing Units and Wildfire Threat within the Wildland Urban Interface
- 11.4 Number and Percentage of Communities at Risk that are Firewise Communities or Covered by a Community Wildfire Protection Plan

As Chapter 4 of the Assessment states, “Living sustainably in the fire-prone landscapes of California will require broad recognition of the inevitability of fire…” This broad recognition has now surely been achieved, through the extent, intensity, impingement directly into fully developed urban areas, catastrophic loss of life, vast property destruction, and massive air quality impacts of recent wildlife seasons. Persons in California and beyond have become more connected than ever to the role that wildfire plays in our forests, wildland urban interface areas, and urban areas themselves. While this is not the most positive way for connecting people to forests and trees, it is nonetheless very real and very present today.

Prior to the arrival of predominantly European settlers, Native Americans located in what is now the State of California had an intimate relationship with the area's wildlands. Native Americans intentionally ignited fire on the landscape for various purposes that included maintaining an open landscape free of shrubs, managing wildlife, and favoring certain plants for the construction of baskets and other cultural items. Native Americans in California are working to re-establish this kind of relationship with fire on their remaining tribal lands. CAL FIRE programs and other State programs have been supportive of this activity.

The disruption of fire regimes that began with European conquest has created conditions in many California ecosystems that, in concert with climate change and people on the landscape, are manifesting themselves in the form of increased wildfire activity and severity, with ecological, economic and human consequences. To help wide-ranging audiences better understand wildfire and the current conditions of forests that are conducive thereto, the concept and indicator of Fire Threat (i4.2) has been developed to provide a more human-centric measure of fuel conditions and fire potential in the ecosystem, representing the relative likelihood of "damaging" or difficult to control wildfire occurring for a given area. In California, a large percentage of wildfire ignitions (85%) are human caused, thus the more that people come to understand the relationship between their actions and the fire threat status of our forests, the better people will be prepared to help avoid ignitions, plan for actions that reduce fire threat, and take steps to prepare for and evacuate when needed.

Unfortunately, significant obstacles continue to hinder increases in pace and scale of the fuels treatments needed to reduce fire threat, particularly prescribed fire. Risk aversion to escaped prescribed fire prevents many potential projects, and appropriate weather for safe burning is seasonally limited. Smoke management remains a substantial hurdle, as near-term impacts to air quality from prescribed fire or managed wildfire are weighed against long-term risks from wildfire. Implementation of managed wildfire policies remains problematic, particularly for CAL FIRE, whose primary fire management responsibility is on private lands where fire protection for people and property remains the top priority.

Development patterns have created a fire environment where about 3 million housing units are within Fire Hazard Severity Zones (FHSZ) and are potentially at risk (i11.2). This includes 2.2 million housing units within the Wildland Urban Interface. Thus, land use and community planning, reduction of structure vulnerability, and fire prevention education have increasingly come to the forefront of efforts to reduce the societal impacts of wildfire.

The National Cohesive Wildland Fire Management Strategy includes a goal of creating Fire Adapted Communities, which recognizes the importance of various programs and actions such as community planning, land use planning, education programs, and homeowner responsibility. Two ways this goal can be accomplished are by creating a Community Wildfire Protection Plan (CWPP), or by becoming a Firewise community. Currently, of 1,338 communities identified as Communities at Risk (CAR), 66% (881) are covered by a CWPP (individual, regional or countywide) and/or are recognized by the Firewise program (i11.4). Numerous other communities are at various stages of CWPP development.

The CAL FIRE Land Use Planning program works with local government to address wildfire risk as part of the safety element in city and county general plans, as required in Government Code 65302. Additional components of community safety include education programs such as “Ready, Set, Go!,” and homeowners taking responsibility to reduce their risk.

A significant component of community safety is homeowners acting to reduce their individual wildland fire risk. CAL FIRE provides education and assistance to homeowners in this effort, including through defensible space home inspections that verify homeowners comply with regulations related to establishing a defensible space and reduced fuel zone, clearance around propane tanks, adequate display of address numbers, and proper configuration of chimney and stove openings.

Relevant Opportunities from Chapters 4 and 11 of the Assessment are:

- Enhance understanding among the public, land managers and fire agencies that wildfire is endemic to much of California, that it is inevitable, and that adverse impacts can be strategically addressed through a variety of programs and activities.
- Emphasize landscape-level planning consistent with typical modern-era fire events. This requires cross-agency/landowner coordination, and consideration of multiple objectives.
- Continue state and federal efforts to engage in local land-use planning to assure that fire risk concerns are understood and mitigated when planning future development.
- Promote sound planning and mitigation in the wildland-urban interface to make communities more resistant to damage from wildfire.
- Incentivize and provide financial and logistical support for individual landowner and homeowner efforts to mitigate fire risk through fuel management, structural retrofit, and engagement in community wildfire planning efforts.
- Promote wildfire planning, education, safety, response, and insurance programs that better recognize homeowner and community efforts to mitigate wildfire risk.
- Continue to support community involvement in developing Community Wildfire Protection Plans (CWPPS), and becoming Firewise and/or Fire-Adapted Communities.
- Continue the CAL FIRE Land Use Planning program to work with local government to address wildfire risk as part of the safety element in city and county general plans, as required in government code 65302.
- Continue to support programs such as “Ready, Set, Go!” to educate landowners in wildfire preparedness, and encourage them to take responsibility for their home and community.
- Continue and improve the CAL FIRE defensible space inspection program to assist homeowners in correcting problems that could put them at risk from wildfire.

Section 2: Connecting Children with Trees and Forests through Outdoor Education

Spending time in nature, or in natural environments such as among trees in city parks, has numerous documented beneficial effects. Among them are increased health and vitality, relaxation and rejuvenation, and a stronger feeling of connectedness to the natural world. People who spend time outdoors in nature also can come to appreciate the various efforts and programs that provide and maintain natural spaces and urban forests, and become strong supporters and advocates for them.

The benefits of environmental and outdoor education to children have been well-documented in the academic literature. Children are particularly sensitive to their surroundings, and extended experiences in the outdoors in nature and natural settings can help them with their physical and emotional development. They learn the importance of forests and natural landscapes to the functioning of ecosystems and the biosphere. Such activities, especially for children in urban areas, help to counter what is known as "nature deficit disorder" (a severe lack of exposure to natural systems and spaces) (Warber, et al., 2015).

Program representatives we interviewed spoke of the many children who had never before seen forests, forest fauna and flora, or even native Californian ecosystems, and how they were completely enthralled with their first encounters. Some discussed how these programs had made children from strictly urban environments want to pursue careers in the out-of-doors and nature conservation. The academic literature also demonstrates how outdoor education can assist children, who have experienced difficulty learning in traditional classroom environments, thrive in an outdoor classroom environment and model (*CN). The lasting impacts and benefits of outdoor education on children can be substantial.

In a time of increasing awareness of the importance of connecting children to nature, the significance of outdoor education program opportunities is clear. But the programs faced many ongoing challenges, even prior to the COVID-19 debacle. These have included diminishing access of communities to natural spaces and park settings, increasing logistical costs such as for transportation, competition for limited space, and liability concerns. All of these have been negatively impacting communities, particularly in lower income urban areas across California.

Prior to the COVID-19 outbreak of 2020, there had been a modest decrease in in participation in many outdoor education programs for school-age children. Moreover, since 2005 the overall network of outdoor education facilities offering such programs has been slightly decreasing, according to a recent report by the Lawrence Hall of Science at UC Berkeley *(CN). This section offers a brief examination trends of student participation in outdoor education over the last decade in California, as it relates to connecting them to forests and rangelands, and the future of outdoor education programs and facilities in California in a post-COVID-19 world.

<u>Trends in Outdoor Education Program Through early 2020</u>

To ascertain trends in outdoor education programs in the state, we reached out to a significant number of program coordinators, school administrators, and others. The information in this section and the discussion summarizes the overall picture that emerged from these conversations. Given the lack of any central database or repository of such information, putting the composite picture together was a challenge.

Our focus here is narrowly on the numbers of student participants in those programs that provide environmental education and related experiences to students in natural landscapes over the course of 2+ days, i.e., where they have at least one overnight in an outdoor (or small cabin) setting. Program outings of single-day (or shorter) local field trips with similar agendas were not included, and this removed a significant proportion of school-related outdoor person-days from consideration. While the latter also clearly benefit children in connecting to nature, they would have been much more difficult to quantify and summarize.

We also did not include multiday outdoor camps where environmental education is not the primary focus of the programs. These include the many camps sponsored by various religious groups. Private associations that provide opportunities for children to connect to the outdoors, such as the Boy Scouts and Girl Scouts of America, were also largely excluded from our findings, primarily because of the difficulty of obtaining comprehensive regional data and trends for California.

The outdoor education programs/facilities we looked at can be classified by their main sources of funding – public, private, mixed, and unknown. Of the four, programs receiving both public and private funding account for about 91% of the students served, according to the information we received. Others had served far fewer students (9% of the total, on average).

Figure 1 shows the number of student participants by program funding class from 2010 to 2019. The graphic shows a fairly stable number of between 110,000 and 120,000 total students annually through the period, with a slight downward trend from 2015 through 2019. Unfortunately, the recent effects of COVID-19 pandemic restrictions on these numbers in 2020, not included in the graphic, are reported to have been catastrophic.

For a number of years, prior to the COVID-19 pandemic, multi-day outdoor education programs had reportedly become increasingly in demand by schools across the state. However, the downward trend from about 2015 on, according to sources, may be attributed in part to an increased wariness and sensitivity among the public to the hazards of transporting students long distances (e.g., to outdoor education venues). In April of 2014, in a widely publicized and horrific accident, a bus carrying students home from a trip collided with a FedEX tractor-trailer on Interstate 5 near Orland, CA. The collision and fire killed seven students from across several high schools in southern California, killed three chaperone adults, and injured dozens more.

An additional concern has been the recent severity and wide scope of wildfires and the prodigious amounts of smoke they have produced. Parents may be reluctant to send their children to outdoor camping venues for concern that they may be adversely affected by the wildfires, either directly or indirectly. Still, the overall amount of positive feelings towards outdoor schools has remained high, and these numbers demonstrate that even given all the serious concerns, at least up until 2020 many parents were wanting to send their students to outdoor school opportunities

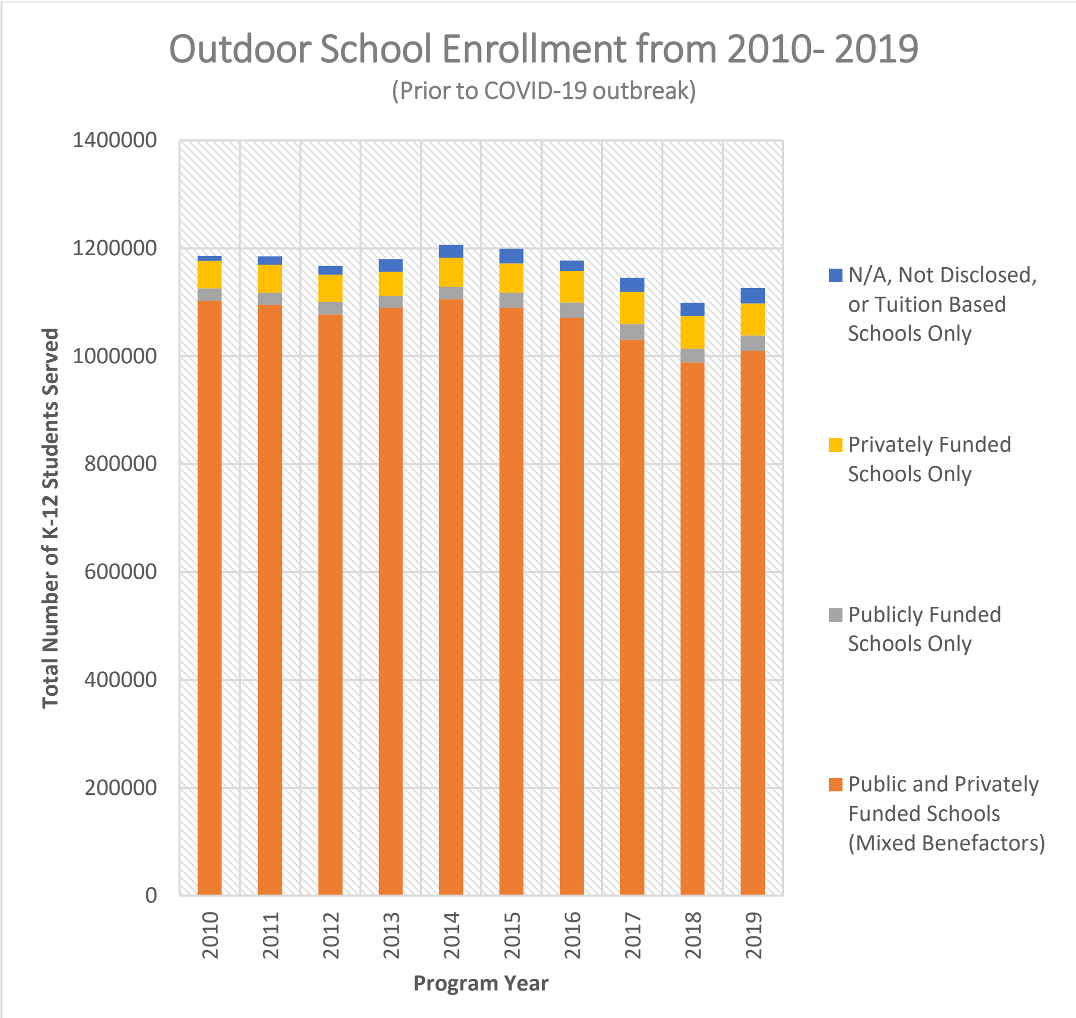


Figure 1. Number of K-12 Students Served in California by Year and School Type, 2010-2019.

Attributing general locations of home residences of student participating in outdoor education venues was also a challenge. In many cases, outdoor schools did not have such detailed information available for their clients. Thus, given the information we obtained, it was not possible to determine which communities across the state were well or poorly served in this regard. Nonetheless, we attempt to approximate the counties of residence of the children participating in the multi-day programs, based on the more general information provided to us (Figure 2).

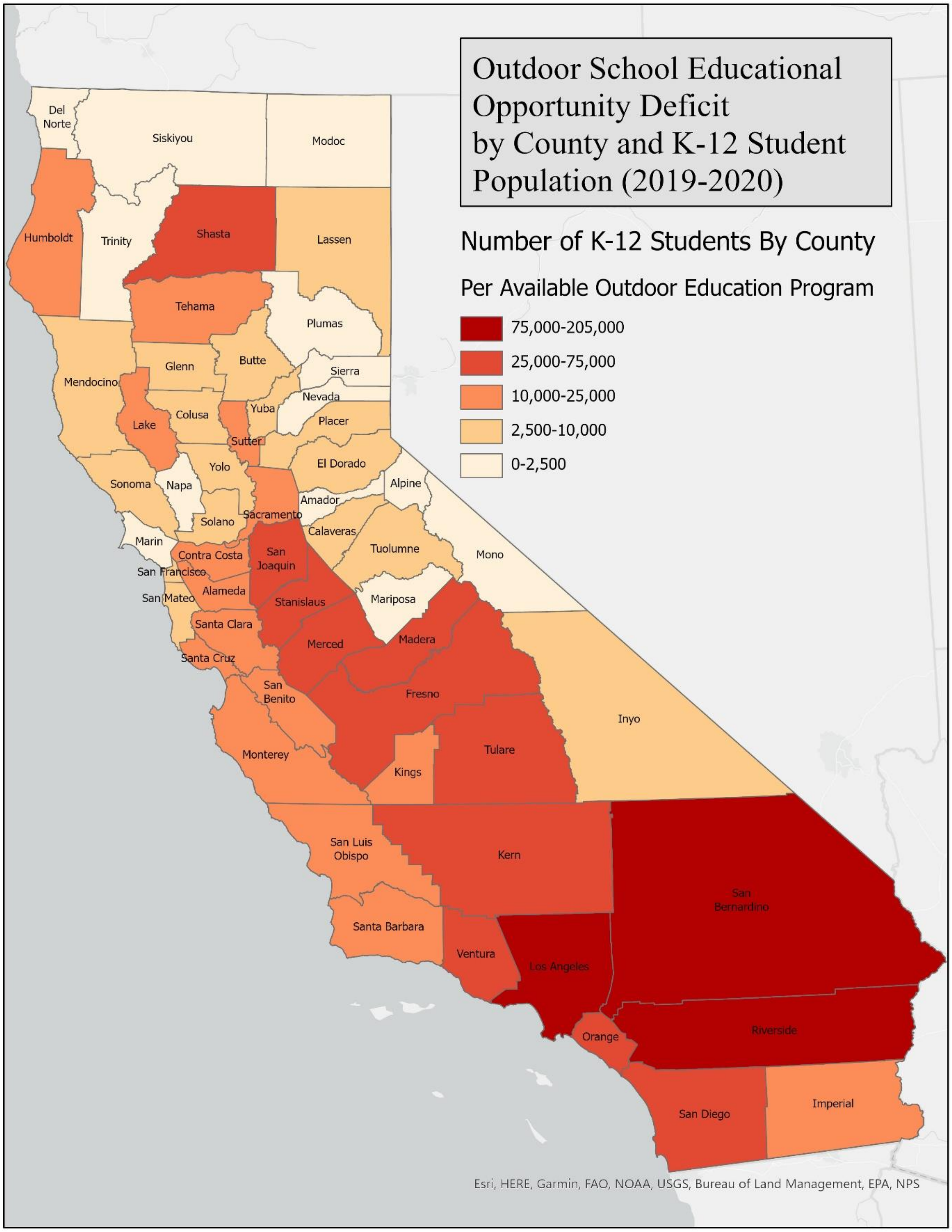


Figure 2. Estimated opportunity deficits for student participation in outdoor education programs.

Our data, with its acknowledged limitations, appear to show that children in southern California have considerably less access to outdoor education programs than those in other areas of the state. San Bernardino County is far and away the most affected in this analysis, but Riverside and Los Angeles counties also appear at the top of the deficit list. In contrast, the large urban areas of northern California (the Bay Area and Sacramento) appear to have a higher proportion of their students with access to these programs.

Discussion

Accumulating and assessing data on outdoor education remains a challenge, as there is no central curated database for the state. Few reports similar to this chapter appear to have been conducted, that describe of the status of outdoor education industry in California. Nonetheless, the many organizations that responded showed an eagerness to assist us in our data collection efforts.

We identified 179 organizations recommended by the Association of Environmental and Outdoor Education (AEOE -- an affiliate of the North American Association for Environmental Education), the Children and Nature Network, and the Outdoors Empowered Network. We then directly contacted each organization that fit the parameters of the survey; each group was emailed multiple times and called multiple times to ensure a chance to participate. We chose groups that brought children outside of the traditional classroom to a natural forest, an urban forest, and/or a rangeland area to learn about natural sciences, ecology, and forests. The main focus and criteria were that they had to have K-12 groups travel to outdoor areas that were centered around interactive learning and educational experiences in forests, urban forests, and/or rangelands.

Around 39% of the groups contacted, or 69 organizations, provided data or program information. Some other organizations that were contacted were not able to respond or provide data due to programs being suspended due to COVID-19, financial challenges, diminished staff, or other logistical reasons. The AEOE, Children and Nature Network, and the Outdoors Empowered Network proved invaluable in our search for these groups. We also received assistance from groups affiliated with University of California–Davis and the California Department of Education.

The direct or indirect impact of wildfires on outdoor schools has also affected these programs. At least 3 of the schools contacted that were not able to respond were affected directly by a wildfire in 2020, and unfortunately at least 2 of those schools suffered severe damage to their facilities. There were also multiple schools that had been either repairing facilities from wildfires from the years 2015 to early 2020, or they had extremely close calls with wildfire events this year and in previous years.

Of the approximately 39% of groups that responded to our requests for data, the vast majority were able to share enrollment data or other insightful information about their programs. Many of the programs shared very similar trends and challenges. These programs on average had seen increases in enrollment up until the COVID-19 outbreak, with some modest decreases in the last two years. Some programs were able to increase their camp infrastructure as well prior to COVID-19, with a majority experiencing positive financial growth.

Present and Future Challenges for Outdoor Education

As previously noted, there has been a growing list of challenges for outdoor education, even prior to the COVID-19 outbreak. Many programs were forced to close their doors for spring and for their summer seasons of 2020 to address mandates set forth by the state and local governments for mitigating COVID-19. Many schools across the state abruptly switched to virtual learning programs, which in turn led to some outdoor education programs switching their strategies and curriculum material overnight. Some programs were not able to adapt fast enough and made the difficult decision to cancel their summer season altogether, in the process losing out on arguably the most important time of year economically for outdoor educational programs.

An additional concern is that many programs face severe financial and public health uncertainties for their upcoming seasons in 2021. A large majority of programs FRAP interviewed emphasized the fact that their programs need to plan months or even up to a year ahead of every school semester season or summer season. With cases of COVID-19 surging in many places across the state as of the writing of this report, many programs are seriously concerned about closing for a subsequent season while trying to remain afloat economically.

Also expressed by those interviewed were serious concerns regarding meeting the increased demand but decreasing availability of program space for K-12 groups. While it is encouraging to see areas of growth in these programs, there were some, particularly in larger metropolitan areas like Los Angeles and parts of the Bay Area, that for the last couple of years were forced to increase the size of their waitlists for K-12 classes. Therefore, the demand for outdoor education has started to surpass the capacity many outdoor schools can handle.

Significant financial difficulties and logistical challenges that have been affecting the outdoor education industry for some time, not just in California but across the country. Programs from all over California reiterated the fact in our interviews and survey that transportation costs had always posed challenges. This would, every year, force some schools and school districts from lower-income areas to have to limit the number of children sent to these outdoor schools, or cancel their partnerships with the outdoor schools altogether to save money. This was a common issue not just for publicly funded outdoor schools, but also for many private and NGO outdoor schools. While some schools have been able to subsidize these costs and allow children from lower income areas to travel to outdoor schools, the effort can entail several years. It can require a well-coordinated effort that many teachers and schools do not have the resources or time to accomplish. Schools typically do everything in their power to send as many students as possible to these programs, but if there isn’t financial support, schools and school districts must make difficult choices in order to stay afloat economically.

As briefly noted above, many schools are now being directly or indirectly affected by severe wildfires. These schools have been under the increasing threat of wildfires for the last ten years. The 2020 fire season was the worst fire season on record in California, with many of the fires presenting serious dangers to the outdoor school facilities, students, and personnel. Some schools interviewed for this report were fortunate to remain unscathed, partially due to beneficial forest management surrounding their properties. Other schools and organizations were not so fortunate, and received devastating blows to their facilities. The facilities of one group interviewed had been almost totally destroyed in 2018 by a severe wildfire, and was about to resume construction when the COVID-19 outbreak started. Even as of the writing of this report, another school was severely affected and damaged by the Silverado Fire outside of Irvine, California, with the smoke from that fire impacting many of the school districts and neighborhoods that the school has served for decades

Outdoor education, and environmental education in general in California, needs more financial support more than ever to meet increasing costs and levels of demand. At the federal level, the Environmental Protection Agency and the Department of Education have provided grants and financial support in the past. There is a great opportunity for the departments within California's state government to help finance these schools. Some of the schools are fortunate enough to receive public funding and receive private funding. However, many more grants and assistance are required to help these programs survive through economic and logistical hardships presented by COVID-19. By investing in these programs, the state will increase its educational opportunity goal attainment, as well as help spur new opportunities for interactive learning through outdoor education. One of the state's goals pertaining to education is to promote environmental literacy as well as promote Next Generation Science Standards. These outdoor educational programs are well equipped to adapt to this curriculum, and some are actively incorporating fire science and forest science by including curriculum from Project Learning Tree and Fireworks, both federal curriculum material and programs for forestry and fire science respectively. With financial support, these programs will bring California much needed advancement of educational goals, while also providing new ways to teach children through these challenging times.

Section 3: Trends in Publicly Accessible Forests and Open Space

Proximity to nature, in larger parks, forests and open space is essential for providing opportunities for people to get out and enjoy their benefits. In this brief section, we look at the status and trends of natural open space that is publicly accessible, on a county basis.

The Importance of Accessible Natural Open Space

One measure of progress towards connecting people with trees and natural open space is the increase over time in the space available in a given locale, region or county to have experiences in nature. These areas can come in the form of large city parks, working rangelands and forests, and significant reserves of natural landscapes.

A number of efforts of private organizations and government agencies have been underway for years to acquire private land in forests and rangeland, and in many cases make it available for hiking, biking and other public uses. One indication of the level of effort in this regard is the number of land trusts active in various parts of California. More than 200 land trust organizations of various sizes are operating in California, with their primary mission of conserving land in at least an undeveloped, semi-natural state, and other open space. However, most of the land conserved by these organizations has been through the purchase of conservation easements, not fee title (i.e., outright ownership), and due to issues involving privacy, liability, lack of infrastructure, and other concerns, has not typically been made accessible to the public.

We used a GIS layer, published annually by Greeninfo.net, of protected areas of the state, called the California Protected Areas Database (CPAD). (They also publish a parallel GIS layer of all lands under conservation easements, called the California Conservation Easement Database (CCED) which was not used in this study). The job of tracking the details of conservation status for about 100 million acres of the state on a yearly basis has been a challenge, but since the original releases in the early 2000s the newer editions of the data have become ever more accurate and comprehensive.

Over the past decade or so, the CPAD have shown consistency in reporting the locations and number of open space acres in the state conserved over time. For this report, we used CPAD from 2012 to 2020 to track trends in the acquisition of natural lands, focusing on those that are made accessible to the public in the form of parks and open space. We used counties as the basis for tracking these trends. Contiguous tracts of land 40 acres and greater were included in this analysis.

Clearly, counties with significant amounts of acreage in National Forests, National Parks, or in lands managed by the Bureau of Land Management have a huge advantage in publicly accessible open space over those without. These are counties located mainly in mountainous regions of the Klamath, Cascades, Transverse Ranges, and Sierra Nevada, as well as those with large areas of the Mojave and Sonoran deserts.

Nonetheless, we tallied up relative rates of growth at the county level, of accessible natural open space over the 8-year period of data we examined (figure 3). The average rate of areal increase for the 58 counties was 2.6%. The minimum observed was 0.05% (Glenn County), while maximum was 11.4% (Placer County) (Figure 3). Other counties with small percentage increases had such due to the presence of large tracts of accessible public lands, whereas other notable counties with substantial increases (and few areas of public land) were Solano, San Francisco and Sacramento counties.

Discussion

The rate of growth in publicly accessible natural landscapes appears to have been fairly modest over nearly the past decade, at an average rate of about 0.3% per year (by county). The high cost of land in the state, particularly in areas closer to metropolitan centers, has made larger land purchases for conservation and public use in many cases prohibitively expensive. Acquisition of land for conservation via other means (e.g., conservation easements) has been proceeding much more quickly, likely due to the much lower costs per acre associated with such transactions. But these latter lands most often do not become open to public use.

Some counties, recognizing a strong need for more accessible open space, have established special districts primarily for acquiring such lands. An example is Sonoma County's Agricultural Preservation and Open Space District, funded by a small county sales tax. Active since 1990, they have secured agreements on more than 180 separate land parcels in the county. Even so, Sonoma County accessible open space has only increased an average amount over the last 8 years (2.7%).

The presence of natural open space within a county, in the form of forests and rangelands, while a prerequisite to their public use does not guarantee that such areas will be effectively used to help connect people with trees and nature. In southern California, there are large expanses of deserts, mountain forests, and chaparral, but most are located away from most people and require means of transportation to get there.

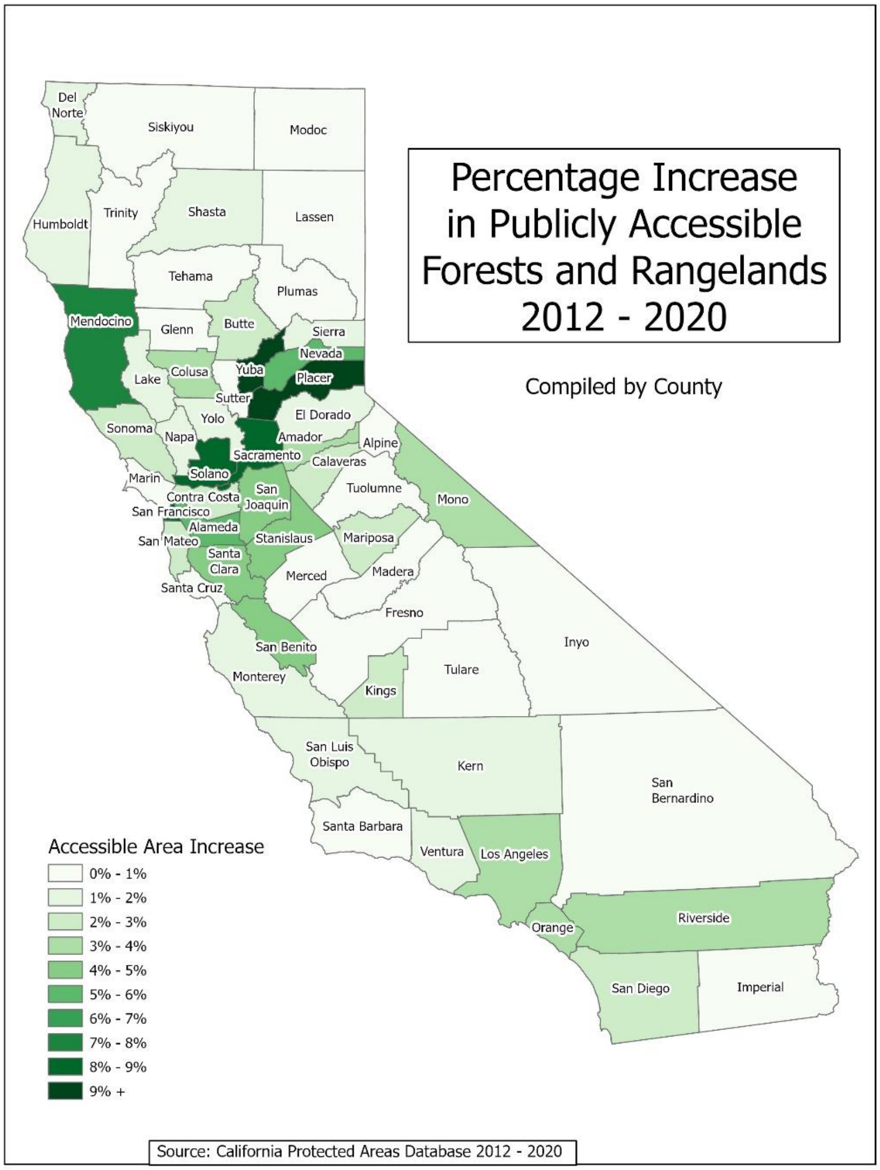


Figure 3. Counties with 8-year percentage increase in accessible parks (2012-2020), comprised of forest and rangelands greater than or equal to 40 acres

Summary

National Priority 3.6 from the Forest Service's Redesign framework stresses the importance of connecting people with trees and natural open space. In section 1, we have shown how various parts of the Assessment relate to CAL FIRE's mission in protecting natural landscapes, and with that their availability and importance to people. A number of CAL FIRE programs thus relate to this framework goal, as do many other State programs that are touched on in the Strategy Report.

In section 2 we reported the status of outdoor education programs (primarily prior to the COVID-19 pandemic) that connect children to the out-of-doors. These programs were often struggling, but were relatively stable prior to the pandemic. They may not have been serving economically disadvantaged communities, especially in southern California, as well as some other regions of the state. Very unfortunately, the pandemic has had an effect on the programs that can best be described as devastating. Many will require significant infusions of funding to restart, if that is even possible.

Section 3 gave a brief overview of recent trends in the acquisition of publicly accessible natural lands, on a county basis. For the most part, higher percentage of additions have been in counties in northern California from San Francisco and Solano counties to Placer County. And smaller, more developed and urbanized counties are faring generally better than larger, more rural ones. This may be a positive trend in that the additional lands being acquired are more proximate to urban populations where they are more needed.

California Forest Action Plan Update 2020 Addendum:

Bordering States Issues

Key Findings

- As forest and rangeland fire has become more frequent, larger, intense, and damaging in the Western United States, both direct (areas burnings across state lines) and indirect effects (e.g., vast amounts of smoke travelling long distances to affect rural and urban areas alike) have become more problematic at the multi-state level. Two examples along the California-Oregon border include the almost-500,000-acre Biscuit Fire in 2002 and the 157,000-acre Slater/Devil Fires in 2020. Thus, efforts of states to jointly assess wildfire hazard along their borders have become increasingly important. [Coordinated Fire Hazard Mapping (Oregon)]
- Efforts to fully understand, quantify, and model forest carbon stocks and fluxes and biogenic carbon embodied in forest products are still in relatively early phases. Opportunities to increase our understanding of these matters can lead to important findings for understanding climate change and considering related policies for mitigation and adaptation. Given the similarities across Pacific Coast temperate forests and the tight interrelationships among this region's forest product flows, enhanced understanding of these carbon issues is important for policy development for forest management and wood products markets. [Pacific Coast Climate Change and Forest Carbon Initiatives (Oregon, Washington, and British Columbia)]
- Forest resilience concerns, including wildfire and forest-pest-related mortality, increase with the drought conditions and warming temperature trends we are already seeing due to climate change. Areas of concern include densely populated areas in the wildland urban interface (WUI), such as the Lake Tahoe area, as well as wildland areas. [Tahoe-Central Sierra Initiative (Nevada)]
- Movement of damaging insects and diseases across state or national borders is a critical concern for the health of our forests (wildland and urban) and rangelands. Non-native pests have a long history of causing severe damage to California forests. The potential for new damaging pests to arrive and become established is great. [Exotic insect and disease threats (Oregon, Nevada, Arizona, International)]
- Wildlife habitat loss and degradation on rangelands can occur due to development, invasive species, interruption of natural disturbance processes such as fire, and agricultural land uses such as intensive grazing. The consequences of habitat loss can be significant for individual species and ecosystems. [Greater sage-grouse (Nevada)]

Introduction

This chapter discusses important forest and rangeland issues and activities that are occurring along California's borders with Oregon, Nevada, and Arizona. International connections also are important, particularly as potential sources of new forest pests. Because of its case-study approach, this chapter does not directly incorporate Montreal Process Indicators. Four of the five case studies are priority areas for CAL FIRE. The fifth (greater sage-grouse) is provided to illustrate how important work extends to rangelands as well as forests. Several common themes are identified across the case studies, including the importance of good monitoring data, use of the best science, and the value of collaboration.

Following the presentation of key assessment findings, this chapter starts with detailed background information on the forest and other vegetation types along the interstate borders, then continues with five case studies involving valuable collaboration to address forest and rangeland problems and opportunities:

In this chapter, we review six case studies of collaboration across state and international borders. The areas that are of high priority for CAL FIRE are identified in **bold typeface**

- **Coordinated Fire Hazard Mapping (Oregon)**
- **Pacific Coast Climate Change and Forest Carbon Initiatives (Oregon, Washington, and British Columbia)**
- **Tahoe-Central Sierra Initiative (Nevada)**
- **Invasive Pest Protection (Oregon, Nevada, Arizona, other states, and internationally)**
- Greater Sage-Grouse Habitat Protection and Enhancement (Nevada)

Finally, the chapter concludes with a discussion of the lessons that can be learned from these vase studies.

Bordering States Area Background

California shares extensive borders with Oregon (350 km; 217.5 miles) to the north, and Nevada (985 km; 612 miles) and Arizona (386 km; 240 miles) to the east. In this section, we describe and quantify the general vegetation types that span these borders, with some implications on cross-border management and their relevance to the Forest Action Plan.

Omernik et al. (2011) developed an ecoregion-based map of the United States that has been used by the US Environmental Protection Agency, the US Geological Survey, the Natural Resources Conservation Service, and the US Fish and Wildlife Service, among others. It has regions similar to the Bailey's Ecoregions used by the US Forest Service, which it employed as source material, but it provides more floristically-based units and descriptions well-suited for our purposes.

A full treatment of Omernik's work is beyond our focus here. The level III hierarchy of Ecoregions divides the state into 13 regions which resemble Bailey's Ecological Sections. Eight of these Ecoregions span California's borders with other states (Table 1).

Table 1: Overview of Ecoregion Types for Bordering Area.

Omernik Ecoregion Name	State(s) shared with
• Coast Range	OR
• Sierra Nevada	NV
• Eastern Cascades Slopes and Foothills	OR
• Central Basin and Range	NV
• Mojave Basin and Range	NV, AZ
• Klamath Mountains/California High North Coast Range	OR
• Northern Basin and Range	OR, NV
• Sonoran Basin and Range	AZ

Omernik provides a level IV schema, which relates more closely to generalized vegetation units. Ecological units that are shared with bordering states are shown in Figure 1. More details are provided below, by state.

Oregon

Oregon has the shortest border with California among the US states, with about 350 km. However, it shares by far the largest area of forest management opportunities with our state. Almost 54% of the border with California is comprised of vegetation types of high priority to the Forest Action Plan. The main types in this category are: Border High Siskiyous; Fremont Pine/Fir Forest; Inland Siskiyous; Klamath River Ridges; Southern Cascade Slopes; Warner Mountains; and the Northern Franciscan Redwood Forest. Together, the former comprise almost 188 km (54%) of the OR-CA border (Table 2).

A wide variety of tree and forest types span the border with Oregon. Main forest types include Klamath mixed conifer, true fir (*Abies* sp.), open eastside and serpentine soil pine (ponderosa, Jeffrey, western white), and in the coastal area, coast redwood. Areas with western juniper as the dominant tree species were considered of limited forest management value.

Oregon has experienced the most cross-border wildfires with California, due to high wildland fuel loadings, and common wildfire ignitions. At the time of this writing, the Slater Fire this fire season burned 157,270 acres along the CA-OR border, primarily in the types Border High Siskiyous and Inland Siskyous. The site of ignition was in California, but it moved northward into Oregon.. Historically, the huge Biscuit Fire of 2002 burned in both states, although nearly all the area was in Oregon. The Biscuit Fire burned nearly a half million acres of forest and shrublands.

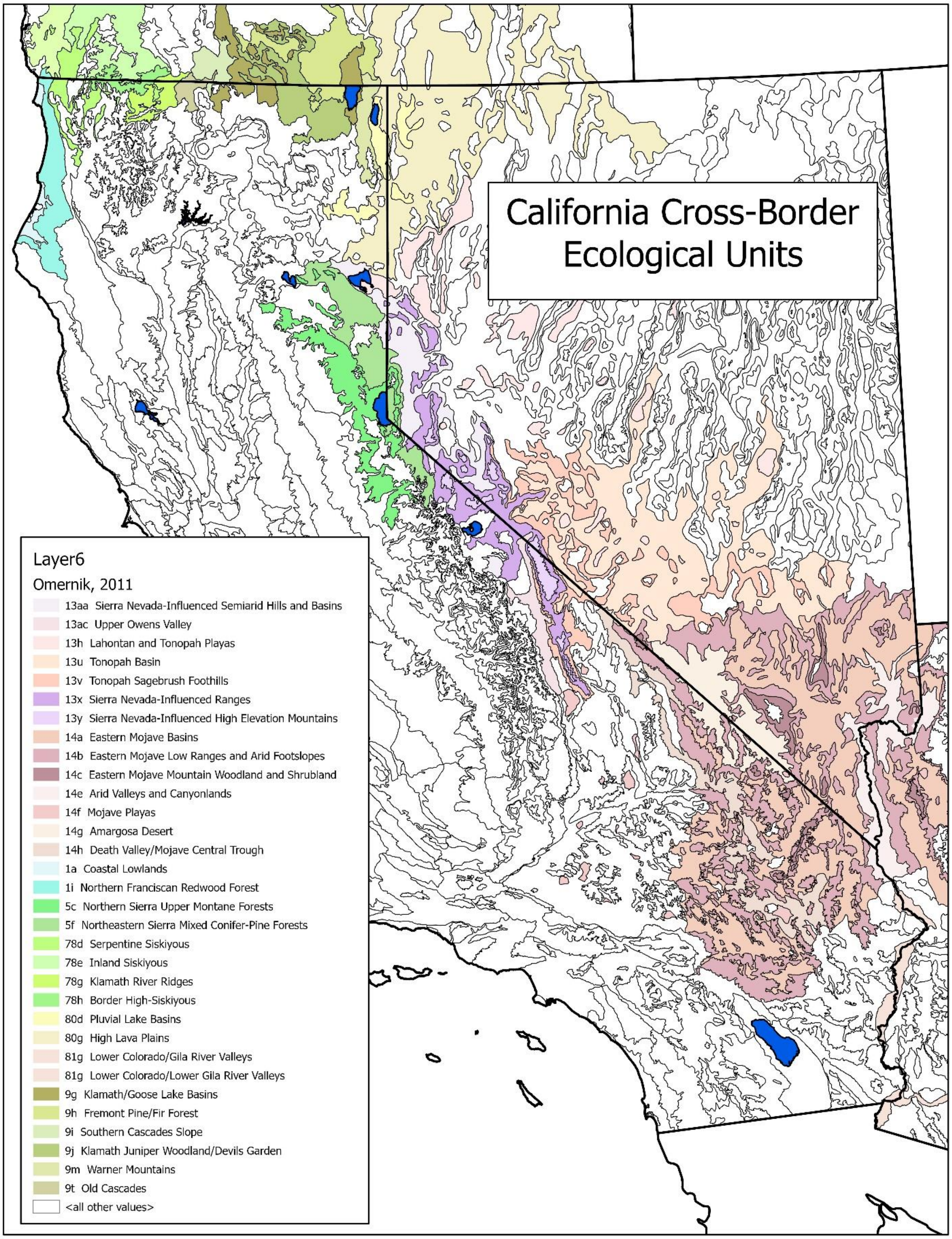

California Cross-Border
Ecological Units
Layer6
Omernik, 2011
13aa Sierra Nevada-Influenced Semiarid Hills and Basins
13ac Upper Owens Valley
13h Lahontan and Tonopah Playas
13u Tonopah Basin
13v Tonopah Sagebrush Foothills
13x Sierra Nevada-Influenced Ranges
13y Sierra Nevada-Influenced High Elevation Mountains
14a Eastern Mojave Basins
14b Eastern Mojave Low Ranges and Arid Footslopes
14c Eastern Mojave Mountain Woodland and Shrubland
14e Arid Valleys and Canyonlands
14f Mojave Playas
14g Amargosa Desert
14h Death Valley/Mojave Central Trough
1a Coastal Lowlands
1i Northern Franciscan Redwood Forest
5c Northern Sierra Upper Montane Forests
5f Northeastern Sierra Mixed Conifer-Pine Forests
78d Serpentine Siskiyous
78e Inland Siskiyous
78g Klamath River Ridges
78h Border High-Siskiyous
80d Pluvial Lake Basins
80g High Lava Plains
81g Lower Colorado/Gila River Valleys
81g Lower Colorado/Lower Gila River Valleys
9g Klamath/Goose Lake Basins
9h Fremont Pine/Fir Forest
9i Southern Cascades Slope
9j Klamath Juniper Woodland/Devils Garden
9m Warner Mountains
9t Old Cascades
<all other values>

Figure 1. Ecological Units California shares with Neighboring States (OR, NV, AZ).

Table 2. California-Oregon Border ecoregion types.

Omernik Ecoregions (level III and IV)	Linear Km	CFAP value	
• Coast Range Ecoregion (region 1)			
○ 1a Coastal Lowlands	1.4	Low	
○ 1i Northern Franciscan Redwood Forest	14.2	High	
• Klamath Mountains/California High North Coast Range (region 78)			
○ 78d Serpentine Siskiyous	22.3	Low	
○ 78e Inland Siskiyous		28.9	High
○ 78f Coastal Siskiyous	1.6	High	
○ 78g Klamath River Ridges	25.8	High	
○ 78h Border High Siskiyous		47.9	High
• Eastern Cascades Slopes and Foothills (region 9)			
○ 9g Klamath/Goose Lake Basins	65.6	N/A	
○ 9h Fremont Pine/Fir Forest		34.3	High
○ 9i Southern Cascades Slope		17.7	High
○ 9j Klamath Juniper Woodland/Devils Garden	32.7	Low	
○ 9m Warner Mountains	17.5	High	
○ 9t Old Cascades	33.1	Low	
• Northern Basin and Range (region 80)			
○ 80g High Lava Plains	6.8	N/A	

Nevada

At nearly 1,000 kilometers long, Nevada shares by far the lengthiest border with California. It also shares the most number of cross-border ecoregions, at five. Relatively little of it is comprised of forest vegetation types that offer opportunities for management in the context of the Forest Action Plan per se. However, significant cross-border shrubland areas fall under the purview of CAL FIRE, FRAP, and the Board of Forestry and Fire Protection.

Vegetation types offering the best cross-border forest management potential are the Northern Sierra Upper Montane Forests (5c) and the Northeastern Sierra Mixed Conifer-Pine Forests (5f). Unsurprisingly, these are both located in the region of Lake Tahoe. Taken together, they comprise only about 57 km of common border, or less than 6% of the total California-Nevada border (see figure 1 and table 3).

The Northern Sierra Upper Montane Forests ecoregion ranges in elevation from 6,000 to 8,000 feet, and its forests have a mix of conifers, including red fir, white fir, Jeffrey pine, sugar pine, incense cedar, and some lodgepole pine. Intermixed are areas of quaking aspen groves and mountain meadows. Some montane chaparral also occurs in areas of harsh exposure, repeated fires, and clearcuts.

Table 3. California-Nevada border ecosystem types.

Omernik Ecoregions (level III and IV)	Linear Km	CFAP value
• Sierra Nevada Ecoregion (region 5)		
○ 5c Northern Sierra Upper Montane Forests	13.7	High
○ 5f Northeastern Sierra Mixed Conifer-Pine Forests	42.9	High
○ Northeastern Sierra Lakes (Tahoe open water)	30.7	N/A
• Central Basin and Range (region 13)		
○ 13x Sierra Nevada-Influenced Ranges	161.5	Low
○ 13y Sierra Nevada-Influenced High Elevation Mtns.	29.5	Low
○ 13v Tonopah Sagebrush Foothills	16.8	Low
○ 13u Tonopah Basin	70.0	N/A
○ 13aa Sierra Nevada-Influenced Semiarid Hills & Basins	57.2	N/A
○ 13h Lahontan and Tonopah Playas	19.7	N/A
○ 13ac Upper Owens Valley	6.8	N/A
○ 13w Tonopah Uplands	6.1	N/A
• Mojave Basin and Range (region 14)		
○ 14b Eastern Mojave Low Ranges and Arid Footslopes	79.3	Low
○ 14c Eastern Mojave Mountain Woodland & Shrubland	16.8	Low
○ 14g Amargosa Desert	154.9	N/A
○ 14a Eastern Mojave Basins	50.2	N/A
○ 14e Arid Valleys and Canyonlands	20.1	N/A
○ 14h Death Valley/Mojave Central Trough	12.4	N/A
○ 14f Mojave Playas	3.8	N/A
• Northern Basin and Range (region 80)		
○ 80g High Lava Plains	152.2	N/A
○ 80d Pluvial Lake Basins	40.1	N/A
Total km	985.3	

The Northeastern Sierra Mixed Conifer-Pine Forests ecoregion includes many of the drier eastside forests of the northern Sierra Nevada that occur north of Bridgeport, CA; in the Lake Tahoe area; and to the northern extent of the Sierra near Susanville. These are mid-elevation dry forests, typically between 5,000 and 8,000 feet, with a diverse mix of conifers, such as Jeffrey, ponderosa, and sugar pines; incense cedar; and white fir. The understory can include sagebrush, antelope bitterbrush, and a fire-maintained chaparral component of snowbrush and manzanita.

Several other types held in common offer extremely limited forest management opportunities. They contain small, isolated stands of ponderosa, Jeffrey, lodgepole and western white pine, and one type has scattered bristlecone and limber pine. The major ones, each with 30 or more kilometers, include the Sierra Nevada-Influenced Ranges, Sierra Nevada-Influenced High Elevation Mountains, and the Eastern Mojave Low Ranges and Arid Footslopes (with Joshua Tree Woodland). Other types are either encountered very infrequently along the border, or are desert with no forest management opportunities.

Arizona

Arizona shares about 381 km of common border with California. The two ecological units along it are of low desert, and offer no significant forest management opportunities of natural landscapes.

- 343 km (89.2%) is of type 81g - Sonoran Basin and Range - Lower Colorado River Valley
- 38 km (10.8%) is of type 14e - Mojave Basin and Range - Arid Valleys and Canyonlands

Bordering Vegetation Summary

California's borders with neighboring U S states total more than 1720 kilometers. Only about 14% (245 km) of the total is in vegetation types with significant opportunities for cross-border cooperation, in terms of forest management. Of that total, more than 75% is on the border with the State of Oregon, with the remainder in the greater Tahoe region of the Sierra Nevada shared with the State of Nevada. The remainder (86% of California's borders with its neighboring states), including all its border with the State of Arizona, is in vegetation types with climates generally too warm and arid for the growth of forests. However, it merits noting that Arizona is believed to be the source of the gold-spotted oak borer (*Agrilus auroguttatus*), an aggressive forest pest that was introduced to California in the late 1990s or early 2000s.

CASE STUDIES

Coordinated Fire Hazard Mapping -- Priority Issue

Wildfire planning and management is a critical cross-boundary issue, with both direct impacts (e.g., fires burning across state lines) and indirect effects (e.g., vast amounts of smoke travelling long distances to affect both rural and urban areas). California's wildfire hazard mapping and planning efforts are currently performed for different purposes at varying scales, including local (e.g., CAL FIRE unit fire plans), regional (e.g., National Forest Plans) and statewide (e.g., Fire Hazard Severity Zones, Priority Landscapes). The methods and approaches vary between these efforts, and largely end at the California state border.

Recent advances in wildfire hazard and risk assessment models and methods leverage nationwide input data (e.g., LANDFIRE) to simulate large wildfire occurrence and behavior in a probabilistic manner. Coupled with local resource maps and asset valuations, planners can quantify wildfire hazard, asset exposure and expected value change over broad areas and across boundaries. Several implementations of these approaches have been undertaken with varying scale and focus, including the recently published national maps of Wildfire Risk to Communities. In 2018, the states of Oregon and Washington completed an all-lands, cross-boundary wildfire risk assessment (Pacific Northwest Quantitative Wildfire Risk Assessment). The results of this assessment are available for planners and the public through a web portal, which provides tools for summarizing hazard and risk.

In 2019, the US Forest Service Pacific Southwest Region, with support from CAL FIRE and others, embarked upon a similar all-lands hazard and risk assessment for California, consistent with the methods used by Oregon and Washington. Initial results are expected in Spring 2021, and will provide an opportunity for fire managers and planners from both states to evaluate wildfire hazard and threat to resources based on data analyzed and mapped in a consistent way. These evaluations can be used to more effectively prioritize work and reduce risk to communities and resources near the border.

Pacific Coast Climate Change and Forest Carbon Initiatives (Oregon, Washington, and British Columbia)—Priority Issue

The intersection of forests and climate change is highly complex and of critical importance for the health and resilience of our forests, carbon flux between forests and the atmosphere, and the resulting climate outcomes. Four sub-national entities along the Pacific Coast—California, Oregon, Washington, and British Columbia are working together on these issues via (1) a broad memorandum of understanding (MOU) and (2) a multi-pronged forest and wood product carbon accounting effort.

MOU The Memorandum of Understanding on Pacific Coast Temperate Forests was executed in December 2018 by California, Washington, and British Columbia. Oregon became a signatory in November 2019. The MOU recognizes that the Pacific Coast's temperate forests have the capacity to sequester more carbon per acre than any other forest type around the world. Given the range and value of the ecosystem services they provide, these forests are critical to the economic and social well-being of forest-related communities and economies, locally and regionally. The MOU recognizes the value of the parties working together to better understand forest carbon dynamics and forest responses to climate change. The partners pledge to work together to explore innovations in fuels management, climate-informed reforestation, forest carbon accounting, opportunities for investments on forests and other natural and working lands to increase carbon sequestration, forest resilience, multi-benefit forest uses, and status of natural-resource-dependent communities.

The MOU establishes a model for how we can all benefit from working together to better understand forest carbon dynamics, and for how Pacific Coast temperate forests are responding to climate change throughout the region, using scientific study, adaptive practices, improved data and enhanced data modeling.

Forest and Wood Products Carbon Accounting Consistent with this MOU, the signatories are actively engaged in a multi-pronged effort to better understand and harmonize robust approaches to carbon accounting for forests and forest products, including related economic and social considerations. In addition to the MOU's four signatory jurisdictions, the U.S. Forest Service Pacific Northwest Research Station (PNW Station) is a leading collaborator in this effort.

Accounting for carbon stocks and fluxes in forests and wood products has become increasingly important as the role of forests in the planet's biogenic carbon cycles has become better understood. Accurate accounting and life cycle analysis is technically challenging. Collecting adequate data throughout the carbon life cycle also is challenging and costly. Across the Pacific Coast region, California and British Columbia have played leading roles in forest carbon accounting and assessment. In California, this has been driven by legislative and administrative initiatives such as Assembly Bill (AB) 32, the Global Warming Solutions Act of 2006, AB 1504 (2010), and executive orders. The forest inventory data collected and processed by the Forest Inventory and Analysis program of the Forest Service's PNW Station are critical inputs to these required forest carbon assessments. In British Columbia, the development and application of the Carbon Budget Model of the Canadian Forest Sector has played a very important role in furthering forest and wood product carbon accounting.

The PNW Station has been an important partner with the Board of Forestry and Fire Protection and CAL FIRE in the data analysis required for AB 1504 forest carbon reporting. Under a new agreement (executed in May 2020) between CAL FIRE and PNW, the Station will continue working to produce the next AB 1504 dataset (based on FIA data from 2011-2020), with work products expected to be completed by the end of 2021.

The agreement also provides for an update of logging utilization information (last updated in 2004), using the FIA Program methodology. The University of Montana Bureau of Business and Economic Research is an additional partner for conducting the logging utilization portion of this agreement. Work products for this effort are expected by the end of 2022.

The third component of this agreement calls for PNW to deliver a comprehensive analysis of the Pacific Coast region's (California, Oregon, Washington, and British Columbia) forest carbon stocks and flux. The proposed report will incorporate new regional analysis evaluating the current ecological capacity of these forests to store and sequester carbon. Work products are expected to be completed by mid-2022.

Under a joint venture agreement between PNW and the University of Montana, an analysis of carbon in harvested wood products and the flow of timber throughout the Pacific Coast region and beyond will be completed as an accompanying effort to account for the movement of carbon from the region's forests into storage as wood products. A separate agreement between PNW and the University of California, Berkeley, also brings the latter institution into the research work on regional wood product flow analysis.

Taken together, these multiple regional studies and reports can serve as a baseline to better understand implications of future management scenarios developed through other research initiatives and how decisions in one area of the region may affect forest management, wood utilization, carbon stocks, and carbon fluxes in other areas of the region. By developing a common set of metrics designed to monitor carbon pool attributes, such as changes in the rate of increasing or decreasing carbon stocks, the region can track progress toward specific goals at the state/provincial and regional level, enabling the Pacific Coast States and Province of British Columbia to better work together toward shared forest and harvested wood product carbon goals. This major effort fully reflects the intent of the regional MOU. They are also consistent with the PNW Carbon Research Initiative.

Complementary to these efforts, CAL FIRE recently has received funding and launched efforts to increase the frequency (to once every ten years to once every five years) and intensity of FIA plot measurements. As the responsible entity for FIA in California, the Forest Service's Pacific Northwest Research Station is an important collaborator for this work. This enhance sampling regime will provide more rapid and complete understanding of the increasingly dynamic conditions and trends on California forest lands.

The Memorandum of Understanding on Pacific Coast Temperate Forests and the work being done under this MOU, a significant component of which is described above, demonstrate the value of regional, multiple sub-national entities coming together to collaborate on research, data analysis, and policy assessment. Collaboration at this scale not only better addresses closely-linked regional forest products economies, it also better allows us to address a critical. complex problem that exists at the global scale—climate change. The approach engages multiple stakeholders, provides for the development and application of enhanced scientific understanding, utilizes improved and more consistent datasets across larger areas, and seeks to inform the important policy choices that must be considered and selected as soon as possible to more quickly bend the curve on global greenhouse emissions and atmospheric concentrations.

<u>Tahoe-Central Sierra Initiative (Nevada)—Priority Issue</u>

The Tahoe-Central Sierra Initiative (TCSI) is the first project to be conducted under the Sierra Nevada Conservancy's Watershed Improvement Program. The TCSI is focused on the greater Lake Tahoe area, which is found at the "elbow" in the border between California and Nevada, as shown in Figure 2.

Figure 2. The Tahoe-Central Sierra Initiative area.
(Image courtesy of the Sierra Nevada Conservancy.)

The primary goal of the TCSI is to increase the pace of large-scale restoration to improve forest health and resilience. Composed of partners from the governmental, academic, nonprofit, and private sectors, the collaborative is working to respond to state and federal mandates for greatly increasing the rate of restoration and improving wildfire protection for communities. The major partners comprising the TCSI include:

- USDA Forest Service (Tahoe National Forest, Eldorado National Forest, and Lake Tahoe Basin Management Unit, which includes parts of Nevada)
- Sierra Nevada Conservancy
- California Tahoe Conservancy
- California Department of Forestry and Fire Protection
- The Nature Conservancy
- National Forest Foundation
- University of California
- California Forestry Association
- Local Nevada fire agencies

TSCI is working to develop and execute novel, science-based assessment, planning, project implementation, and monitoring across a 2.4-million-acre forested landscape. Additionally, the collaborative seeks to address critical barriers to increasing the scale of forest management and restoration by increasing the available workforce and forest product production capacity.

Essential elements of this are developing more flexible funding and fostering new markets for biomass and wood products.

A key component of the TCSI is the effort to develop a "Roadmap to Resilience" to provide a science-based and data-driven approach to restoration that can be applied at multiple scales, from the local project to Sierra Nevada wide. The Roadmap to Resilience has four major components or stages:

1. Framework for Defining Landscape Resilience
2. Resource Assessment
3. Blueprint for Restoration
4. Resilience Dashboard

The science-based Framework for Defining Landscape Resilience was developed through a collaborative process and incorporates an initial eight social and ecological values relevant to the Sierra Nevada region (soon to be released is an updated list of ten "resilience pillars," the result of a science-based project funded by CAL FIRE with California Climate Initiatives monies). These values also can be used as indicators for the desired outcomes of the TCSI's restoration projects. The resilience values are:

Ecological Values
- Forest resilience
- Fire dynamics
- Biodiversity conservation
- Water reliability
- Carbon sequestration
- Air quality

Social Values
- Fire-adapted communities
- Economic diversity and social well-being

The Resource Assessment uses the best-available data and science to describe current resource conditions. The gap between current and desired conditions can then be identified, thus showing needed landscape changes. As a part of the assessment, modeling integrates factors such as fire risk, climate impact projections, carbon stocks, wood supply, wildlife habitat, drought tolerance, and water balances. Under this approach, current and projected forest resilience conditions can be compared for multiple potential forest restoration management scenarios.

Based on the outputs of the Resource Assessment, the Blueprint for Restoration incorporates practical limitations (e.g., funding, on-the-ground access) and values (e.g., critical wildlife habitat) to develop an agreed-to vision for a landscape that maintains resilience under the anticipated changing environmental conditions. The Blueprint process also can highlight small-scale, high-outcome opportunities for restoration.

The Resilience Dashboard will allow TCSI partners, agencies, policy-makers, interested members of the public, and others to easily monitor restoration progress and understand how forest health and resilience objectives are being achieved over time.

Five specific forest restoration projects are underway as part of the TCSI:

- Caples Creek Watershed Ecological Restoration Project
- Lake Tahoe West Collaborative Project
- North Yuba Forest Resilience Project
- French Meadows Project
- Sagehen Experimental Forest Project

TCSI partners have thus far garnered $32 million in California Climate Investments grant funds to implement forest health projects. These projects include:

- Thinning 20,000 acres;
- Producing 164,000 (green) tons of biomass; and
- Conducting 8,000 acres of prescribed fire treatments.

These projects will serve to sequester carbon and to reduce wildfire risk, thus helping to protect communities.

The TCSI illustrates the need to skillfully integrate scientific, social, and economic elements into broad-based collaborative processes. Quality biophysical, social, and economic data are needed along with robust-yet-transparent assessment and modelling tools that allow meaningful comparison of the potential outcomes of multiple project alternatives. Once projects are selected and implemented, objective measures of outcomes and related monitoring are needed to evaluate project results relative to intended goals. Strong organizational and collaborative process facilitation skills are needed through the lifespan of these kinds of efforts.

It further needs to be recognized that the current very substantial level of forest health and restoration funding being made available by the State of California is also essential to the success of efforts like TCSI. TSCI's endeavors to foster market-based drivers for forest restoration work can help to ensure that private sector project funding also may be available to support these kinds of projects.

These general lessons, as well as more specific lessons learned from TCSI can be applied broadly throughout the Sierra Nevada and beyond to help increase the pace and scale of forest restoration, the reduction of community wildfire risk, and the enhancement of local economies.

Invasive Pest Protection (Oregon, Nevada, Arizona, Other States, and Internationally)—Priority Issue

Forest insects and diseases do not recognize political boundaries. Therefore, it is imperative that California work with its federal partners and with neighboring states to combat the dangers posed by invasive pests. Protecting California's forests from invasive insects and diseases has become an increasingly difficult task as interstate and international interpenetration have increased over time. Inadvertent movement of forest pests across international borders has become particularly problematic in recent years. The state has expended significant control efforts and suffered great losses of trees due to such introduced diseases as pitch canker (*Fusarium circinatum*), attacking Monterey pine along the Central Coast, and sudden oak death (*Phytophthora ramorum*), weakening and often killing oaks in an expanding range of the moister areas of the state. Other invasive forest and shade tree pests that have become established in California include Port Orford cedar root disease (*Phytophthora lateralis*), South American palm weevil (*Rhynchophorus palmarum*), white pine blister rust (*Cronartium ribicola*), gold-spotted oak borer (*Agrilus auroguttatus*), polyphagous and Kuroshio shot hole borers (*Euwallacea* spp.) and the Mediterranean oak borer (*Xyloborus monographus*). There is constant concern over additional invasive pests arriving from other parts of the U.S.

Addressing forest pest management in California (at the within-state, between-state, and international levels) is strengthened by the pest management programs at CAL FIRE, the Forest Service Pacific-Southwest Region (State and Private Forestry, Forest Health Protection), and the Forest Service national level (Health Assessment and Applied Sciences Team, e.g.). Programs at the U.S. Department of Agriculture, the California Department of Food and Agriculture, and University of California Extension also are very important resources. Collaboration through the California Forest Pest Council, a nongovernmental organization, further strengthens work to address forest pests in the state. For example, the Council publishes the annual California Forest Pest Conditions Report.

Successful pest management programs require a scientifically-based, integrated approach. At the interstate level, firewood moving into California continues to be of concern. For example, firewood brought in from Arizona is thought to have been the source of the gold-spotted oak borer arriving in California a decade or so ago. The California Department of Food and Agriculture (CDFA) keeps a wary eye out for firewood importation (among other things) at the 16 border crossing inspection stations that it operates. Firewood inspected at the CDFA border stations also has been found to be infested with emerald ash borer (*Agrilus planipennis*) and gypsy moth (*Lymantria dispar*), two pests not yet established in California.

The "buy-it-where-you-burn-it" campaign, focused on movement of firewood (by campers, home firewood users, and firewood producers alike), has conducted some very far-reaching and successful messaging within the state and beyond. This campaign originated from The Nature Conservancy's national "Don't Move Firewood" program, was adopted as the "buy-it-where-you-burn-it" campaign in Washington and Oregon, and then instituted in California, in part through the initiative of the Pacific Southwest Region Forest Service Forest Health Program and the California Forest Pest Council. These efforts contributed to the creation of the California Firewood Task Force. The Task Force is entirely voluntary, with minimal funding provided from the Forest Service through grants to CAL FIRE and to CDFA. The "buy-it-where-you-burn-it" campaign is supported by a long list of California and federal land management and pest management agencies.

When a new pest arrives in the state, we often need research to better understand how the pest affects our tree species, how the pest's life cycle manifests itself in our specific biotic and abiotic conditions, how the pest is or is not spreading, what control methods are most successful and cost-effective, etc. For example, when the gold-spotted oak borer appeared in California, CAL FIRE was fortunate to receive a substantial federal American Recovery and Reinvestment Act of 2009 grant to study the species and control methods. Participants in this effort included CAL FIRE, University of California Riverside, University of California Extension, and the Forest Service.

Other exotic pests such as sudden oak death (SOD) have already spread from California to other states, including Oregon (in forests) and Washington (in the nursery trade). SOD is shared across the extreme western part of California's border with Oregon, but that has been true for a decade or so. The disease only recently was found in Del Norte County, which is immediately south of the Oregon state line. The different lineages of SOD disease organisms along the pacific Coast states could pose even greater threats should they come in contact. These states are working together to reduce the potential for such spread and intermingling of diseases.

Danger also exists of invasive pests that have gained a foothold in California moving to our bordering states. The Mediterranean Oak Borer Complex and the Invasive Shot Hole Borer Complexes are established in California and threaten to spread to our neighboring states without work here to suppress or eradicate the pests and conduct education, outreach, trapping and monitoring to control them.

The neighboring states and the federal government work together to survey, trap, monitor and do basic research to minimize the damage caused by invasive forest pests. We work together through organizations such as the California Forest Pest Council and the California Invasive Species Advisory Committee and other west-wide professional and scientific organizations.

In addition to the concern over pests coming across the state line in firewood, our current main challenges with invasive pests and diseases are with other areas (and countries), not so much with our U.S. neighbors. Perhaps the latest example is the Mediterranean oak borer (*Xyleborus monographus,* native to Europe), which has recently been discovered in Napa, Sonoma, Lake and Sacramento Counties and is attacking white oaks [particularly valley oak (*Quercus lobata*) and blue oak (*Quercus douglasii*)]. There is a high likelihood of the expansion of the beetle's range, since it can be moved in infested wood and the females can fly. The pest may be capable of establishing over much of California. Since the Mediterranean oak borer has a preference for attacking weakened trees (e.g., due to drought, fire, SOD, other pests), it may find an ample supply of vulnerable host trees in the state.

A somewhat earlier arrival from Asia are the invasive shot hole borers (ISHB) (*Euwallacea* spp). When boring into a tree, these pests introduce a fungus (*Fusarium* spp.) that causes dieback. The ISHB-*Fusarium* pest-disease complex has been responsible for the death of thousands of trees in Southern California. The complex poses very real threat to the integrity of our urban and natural forests. ISHBs attack a wide variety of tree species, including native species in urban and wildland environments, commonly planted non-native landscape tree species, and agricultural species such as avocados.

CAL FIRE collaborates with federal agencies, such as the USDA Forest Service, the Bureau of Land Management, the National Parks Service, the USDA Animal Plant Health Inspection Service (APHIS) and the US Customs and Border Protection, to protect California from invasive forest insects and diseases. Other state agencies including the California Departments of Food and Agriculture, Fish and Wildlife, and Parks and Recreation and the California Environmental Protection Agency, and others coordinate work on invasive pest issues. Much of the coordination and collaboration occurs through the California Invasive Species Advisory Committee (CISAC) to which all these state and federal agencies belong. Work is also coordinated with the University of California system to meet research needs.

Success in identifying, managing and controlling invasive forest pests requires the collaboration of multiple agencies, organizations, universities, industries and the concerned public. The scope of the issue and the impact of the pests are beyond what any single organization can address. The need to manage the damage and reduce the spread of invasive pests requires scientific knowledge gained from numerous entities both to protect California from pests coming from outside its borders and to protect our neighboring states, the rest of the country and other nations from anything already found within the State. Without management and control, invasive forest pests threaten the health of our forests, woodlands and urban canopies, reduce carbon sequestration, increase wildfire hazards through the production of increased fuel loads, and impact water quality and quantity, air quality, wildlife habitat and recreation.

Greater Sage-Grouse Habitat Protection and Enhancement (Nevada)

Conservation of the greater sage-grouse (*Centrocercus urophasianus*) and its habitat is a concern throughout much of the western United States. California and Nevada are home to populations of the greater sage-grouse. These include a general population, spanning Northeastern California across Northern Nevada (and beyond), and a genetically-distinct bi-state population, straddling the state line from roughly Minden, NV, to Big Pine, CA.

The more northern general population's habitat occupies about 3.2 million acres of rangeland in California and 36.8 million acres in Nevada. The bi-state population's habitat extends across six management units totaling about 4.5 million acres in the two states. In both states, the bulk of the involved lands are federally managed (Forest Service and Bureau of Land Management). Although greater sage-grouse populations have shown long-term decline, the U.S. Fish and Wildlife Service has determined that listing of the species is not warranted, in part because of the collaborative conservation efforts of federal, state, and private partners across the species' range. California and Nevada have both actively participated in conservation of the species. Due to declining populations, all greater sage-grouse hunting has been prohibited by the California Fish and Game Commission since 2017.

State conservation partners include the California Department of Fish and Wildlife and the Nevada Department of Wildlife. The federal partners include:

- U.S. Department of the Interior
- U.S. Department of Fish and Wildlife
- Bureau of Land Management
- U.S. Geological Survey
- U.S. Department of Agriculture
- U.S. Forest Service
- Natural Resources Conservation Service

Other partners have included the Los Angeles Department of Water and Power, Nevada Partners for Conservation and Development, and certain counties.

Invasive plant species and wildfires are the primary threat to greater sage-grouse habitat in the California-Nevada area. Conifer expansion [in particular, western juniper (*Juniperus occidentalis*), Utah juniper (*Juniperus osteosperma*), and single-leaf pinyon pine (*Pinus monophylla*)] resulting from wildfire exclusion, also is negatively affecting the species' sagebrush habitat. The northern California/western Nevada population of greater sage-grouse was significantly impacted by the 316,000-acre Rush Fire in 2012, which burned across Lassen County in California and Washoe County in Nevada. As a part of collaborative efforts, the U.S. Geological Survey is leading ongoing research to better understand the effects of the Rush Fire on the species. Grazing, mining, energy production, and development also are reducing and fragmenting habitat.

California has been an active partner in addressing conservation of the greater sage-grouse. The state has designated the greater sage-grouse as a species of special concern, which requires that it be given a heightened level of consideration in state-mandated environmental review processes. In Nevada, executive orders call for a goal of no net loss of the bird's habitat, and the 2014 Nevada Greater Sage-Grouse Conservation Plan is in place. The latter includes a required compensating mitigation credit program to offset unavoidable impacts to habitat. Further, the state has established Rangeland Fire Protection Associations to strengthen fire protection in rangeland areas.

The Natural Resources Conservation Service's (NRCS) Sage Grouse Initiative and Sage Grouse Initiative 2.0 have been important mechanisms in habitat conservation for the species throughout the West. The NRCS collaborates with private ranchers to develop conservation solutions that benefit both the ranchers and the bird species. In California, conifer removal (primarily western juniper) treatments are planned for priority areas within the Klamath Basin under Sage Grouse Initiative 2.0. The Forest Service and Bureau of Land Management have modified their land use plans in the bi-state region to address threats to the greater sage-grouse. The plans provide varying levels of protection and grazing permit oversight depending upon the designated areas' habitat value for the species.

The Modoc National Forest and Alturas Field Office of the Bureau of Land Management, with collaboration from Modoc County, completed the Sage Steppe Ecosystem Restoration Strategy and related Final Environmental Impact Statement in 2008. The plan covers Forest Service and Bureau of Land Management lands in four California counties (Modoc, Lassen, Shasta, and Siskiyou) and Washoe County, Nevada. There are three California-based Local Area Working Groups (LAWGs), coordinating closely with conservation efforts in Nevada, that guide conservation efforts for greater sage-grouse.

Efforts to address the bi-state distinct population segment are well-organized under the Action Plan for Conservation of the Greater Sage-Grouse Bi-State Distinct Population, which was approved in 2012 by the multi state and federal agency Executive Oversight Committee for Conservation of the Bi-State Greater Sage-Grouse Distinct Population Segment. Recent agency activities under this plan include capturing and moving birds to increase genetic diversity on certain breeding areas (leks), acquiring privately-owned habitat to protect it from development, and vegetation treatments to improve habitat quality. During the 2019 reporting period, the various conservation partners carried out actions to address identified threats to the bi-state greater sage-grouse and their habitats on approximately 13,000 acres in the bi-state area.

The USFWS in 2015 determined that the distinct population segment did not require Federal Endangered Species Act (FESA)protection due to ongoing collaborative conservation efforts. In 2018, a federal court ruled that the USFWS wrongfully denied FESA protection for the bi-state population, and the agency then began a species status review. The California Department of Fish and Wildlife was engaged and provided substantial feedback during this review. In March 2020, the USFWS withdrew a proposed rule to list the bi-state distinct population segment as the result of the substantial conservation efforts carried out by the collaborative team of partners.

The success of state, federal, and local agencies, along with non-governmental organizations and volunteers to protect and improve greater sage-grouse habitat and populations across the California and Nevada border illustrate the power and potential of collaborative efforts. As noted under other bordering-state efforts described in this Addendum, broad-based collaborative efforts that include assessment, monitoring, research, and significant levels of on-the-ground project implementation can achieve significant conservation success. While considered controversial by some, the fact that Federal Endangered Species Act listing for the species has been avoided because of the high level of collaborative conservation efforts speaks to the power and potential of this kind of landscape-level, science-based, multiparty collaboration.

Discussion

The sub-national entities of California, Oregon, Nevada, Washington, and British Columbia and their multiple federal agency partners, and nongovernmental organizations have made significant strides at cross-border collaboration to address challenging forest and rangeland issues that do not recognize political jurisdiction boundaries. The above case studies demonstrate success in several areas (greater sage-grouse conservation, invasive species control, salmonid protection) and offer the promise of future progress in other areas (regional forest and wood products carbon accounting, landscape-level forest restoration, and coordinated fire hazard mapping). It is important for the state, provincial, federal, and local government entities and non-governmental organizations to continue their collaboration on these important conservation issues of mutual concern.

The above case studies consistently highlight what we have learned through collaborative efforts over the past two decades or so:

- Focused, multi-stakeholder engagement is critical to making progress on complex forest and rangeland issues that sometimes require that difficult tradeoffs be made to move forward on improvement or resolution;
- A strong scientific basis with an ecological approach is required to make progress. Sometimes, new scientific understanding must be developed to find solutions. Collaboration in exploring scientific understanding is important to garnering adequate resources for research or data collection and to achieving stakeholder buy-in on findings.
- Better data, sometimes at multiple scales, often is required to fully understand ecosystem issues. Many times, collecting newer and more comprehensive data with shorter refresh cycles is essential to progress in understanding problems, garnering stakeholder agreement on solutions, and meaningfully monitoring outcomes.
- Some problems can only be solved by scaling up in terms of the size of the physical area included, the fields of science included, and the range of stakeholders engaged.
- Agreed-to, scientifically-based monitoring protocols and transparent outcome reporting are essential to determining success or failure, maintaining trust among stakeholders the broader public, and providing a basis for meaningful adaptive management.
- Biophysical, economic, and social factors must all receive significant attention in collaborative process to ensure that a durable and equitable outcome is achieved.

California Forest Action Plan Update 2020 Addendum:

Priority Areas and Regions of California

[Editor's notes:

- *As communicated to the Forest Service, we will be overhauling the Forest Health Grant identified Priority Landscapes described below as:*
    1. *The latest complete wildfire data from 2020 become available in GIS form;*
    2. *Updated spatial data on fire hazard are available in 2021 from a contract.*
- *Once the above are obtained and incorporated into the framework described below, these will significantly change three of the four priority Forest Health landscapes.]*

<u>Overview</u>

Over the past few years, the CAL FIRE Fire and Resource Assessment Program (FRAP) has developed several GIS layers showing the prioritization of landscapes in the State with regard to forest management and wildfire program availability and need. Each GIS layer has been created in the context of forest-related grant programs, such as Forest Health Grants, the state and federal Forest Legacy Programs, and the California Forest Improvement Program (CFIP). Taken together, these priority landscapes serve to geographically define and rank areas for specific programmatic purposes, which are the primary focus of CAL FIRE's forest management efforts.

This multi-part framework for designating priority landscapes reflects the complexity of forest management issues in California and the scope of the several programs that the State has developed in response. These templates are not used as hard-and-fast decision tools, but rather as robust indicative information to inform project selection and funding decisions that typically also involve other considerations, such as landowner interest and capacity, the participation of organized stakeholder groups or nongovernmental organizations, the status of required environmental review and permitting processes, and the source of the project funding.

<u>Forest Health Grant identified Priority Areas</u>

For the submitters of Forest Health Grants, online tools assist in the appraisal of their proposed project areas with respect to four key landscape factors. Two factors are related to reducing local wildfire risk, and two to restoring damaged forests (see https://www.arcgis.com/apps/MapSeries/index.html?appid=f767d3f842fd47f4b35d8557f10387a7) :

- Reducing wildfire risk to:
    - forest ecosystem services
    - communities
- Restoring:
    - pest and drought-damaged areas
    - wildfire-damaged forests

- In addition to the above, forest health grant submitters state whether their proposed projects will have a positive impact on disadvantaged and/or low income communities, and are required to place project locations in the context of communities, particularly those with Firewise and/or Community Wildfire Protection Plans.

The four priority landscapes were developed using a spatial risk assessment approach, where selected assets are evaluated in the context of coincidence levels of threats to those assets. This approach was used extensively in the creation of 22 priority landscapes for the 2010 Forest and Rangeland Assessment (FRAP, 2010). Areas where high-value assets are concentrated with high levels of threats in this system are labeled as the highest priority.

- Reducing wildfire risk to forest ecosystem services [at the Hydrologic Unit Control (HUC) 12 level]
    - Component Assets:
        - Surface water value
        - Carbon (forest) storage
        - Standing timber
        - (Timber) Site quality
        - Large trees (habitat element)
    - Component Threats:
        - Wildfire threat
        - Fire Return Interval Departure (FRID)
    - Highly ranked areas of this PL are concentrated in the central and northern Sierra Nevada, the southeastern Klamath mountains region, and the interior coastal ranges on the west side of the Sacramento Valley [figure 1 (draft); note all figures are grouped at the end of this section]. Note that this information has recently changed significantly due to the extensive wildfires of 2020, and a new version utilizing updated assets (Carbon storage, Standing timber, large trees) as well as updated threats will be needed. This will almost assuredly change the regional prioritizations.
- Reducing wildfire risk to communities
    - Component Assets:
        - Housing, rated by density
    - Component Threats:
        - Fire Hazard Severity Zones
    - Highly ranked areas include the wildland edges around large urbanized metropolitan centers such as across southern California and the Bay Area, the central and northern Sierra Nevada forests, and a large part of the lower elevations of Shasta County [figure 2 (draft)].
    - The same data-currency caveats apply here as to the PL above, but the extensive wildfires of 2020 will likely have less of an impact on the overall pattern of high priority areas.
- Restoring drought and pest-damaged areas (as delineated by the Tree Mortality Task Force)
    - Component Assets:
        - Critical infrastructure

        - Communities
        - Natural resources
    - Component Threats:
        - Levels of forest tree mortality (primarily from the drought of 2012-2016) and presence of hazard trees
    - Given the effects of extensive recent wildfires, highly ranked areas are likely no longer valid in many areas [figure 3 (draft)]. This PL will need to be completely re-done with updated data, post wildfire season this year.
- Restoring wildfire-damaged forests
    - Component Assets
        - Surface water value
        - (Timber) Site quality
    - Component threats
        - (Post-wildfire) Erosion hazard potential
        - High-severity burn wildfire areas
    - The extensive 2020 wildfires have greatly altered the landscapes that went into this prioritization (figure 4 (draft)). In particular, the key high-severity burn threat component will require new information once the wildfire season comes to an end.

Broadly speaking, regions of highly productive conifer and mixed hardwood/conifer forests significantly at risk from wildfire are prioritized in the four GIS landscapes above. These tend to be in the areas of interior regions where forest trees grow the largest and the fastest, in the middle elevations of mountainous areas in the central and northern parts of the state (note: forest trees grow largest and fastest in more coastal watersheds, but the wildfire threat in these tends to be lower). Well-developed (deep) soils, high average precipitation, and warm growing season temperatures all contribute to high rates of tree growth. Thus, the "asset" side of the risk assessments is often highly weighted in these areas. The high carbon loadings resulting from high growth rates also, with time, tend to elevate the potential severity of wildfires, and the threat of widespread insects and disease striking in a low-water year.

In addition to the above GIS priority landscapes, the maps in figure 5.2, on page 130 of the 2017 Forests and Rangelands Assessment, show declared Zones of Infestation (ZOIs) for four major invasive forest pests. These include the pitch canker, sudden oak death (SOD), bark beetles, and the goldspotted oak borer. More recently, the Board of Forestry and Fire Protection also has designated a ZOI for the invasive shothole borer (ISHB). The ZOI declarations by the Board enable authorities to cross onto private property to eradicate the pests. Priority will thus also be given the areas mapped, for efforts to control and mitigate the spread of these forest and woodland pests. Other priority invasive species may be identified over time and CAL FIRE will work with its partners to respond to them accordingly.

California Forest Legacy Priority Areas

The California Forest Legacy Program has recently undergone a major revision to its defining programmatic document, the 2020 Assessment of Need (AON). As part of that revision, a priority landscape was created to reflect areas that would likely benefit the most from program activities (e.g., purchase of conservation easements or land fee title) (figure 5). Note that the federal FLP functions in 34 out of 58 counties statewide. The environmental factors folded into the priority landscape were selected based upon the expert opinions expressed by members of the Forest Stewardship Coordinating Committee (FSCC). Each member was surveyed to indicate the factors most important to the evaluation (at a generalized landscape level) of potential for Forest Legacy programmatic attention.

- Forest Legacy AON Priority Landscape Components, with respective weightings summing to 100:
    - Conversion (to permanent development) threat (19%)
    - Public water supply watersheds (17%)
    - Threatened and endangered species habitat (terrestrial) (12%)
    - Wildlife habitat connectivity (12%)
    - Forest site productivity (12%)
    - Proximity to public or other conserved lands (12%)
    - Priority watersheds for the State and Central Valley Water Projects (10%)
    - Endangered Species Act listed salmonid populations (6%)

A significant difference between the priority landscape results for the Forest Legacy Program and the results for the Forest Health Grant Program is the lack of wildfire threat or aftermath as a component driver. This and other factors tended to elevate areas of highly productive redwood forests closer to the coast in this PL, as well as those forests closer to main urban centers (with the coincident higher threat of development). However, forests on the west side of the central and northern Sierra Nevada and southern Cascades also stand out for their high potential value.

Also unlike some other PLs described in this chapter, the results from the current available analysis is more apt to still be valid, given that most of the phenomena tracked have not been as radically altered on the ground by wildfires in the year 2020.

It is worth noting that the US Forest Service at the federal level currently maintains a public-facing web map viewer which serves the spatial polygon data from all completed Forest Legacy Projects nationwide, including those of California:
https://usfs.maps.arcgis.com/apps/webappviewer/index.html?id=9d083b89bd254c23acf56f8143e0c119

The project shapefile data displayed are those that have been loaded into the federal SMART Forest Service application (on an annual basis) from CAL FIRE's CalMapper database. After it is ingested, our data are then hosted on an ESRI web site, ArcGIS Online. We note that our FLP priority landscape could be added as a layer to this web map application for our state, to show where our analyses indicate additional projects would serve the program needs well.

California Forest Improvement Program Priority Areas

The California Forest Improvement Program recently revised its GIS-based priority landscape, which was first created in 2007 as the Spatial Analysis Project (see https://www.fs.fed.us/na/sap/products/CA/CA-

Methodology.pdf) (figure 6). The original GIS layer was created with a weighted input scheme of 13 environmental factors and other criteria. These included:

- Wildfire hazard
- Private forested land
- Wildland-urban interface and intermix
- Post-fire soil erosion potential
- Recent increased housing density
- Forest industry land
- Forest type
- Priority watersheds
- Areas of forest subject to insects and disease
- Riparian zones
- Threatened and endangered species critical habitat
- Public water supply watersheds
- Proximity to public lands

While the initial SAP priority criteria were quite broad, recent refinement of the top priority areas was done to highlight those areas most in need of program funding. Unlike for the Forest Legacy Program priority landscape, the SAP map results include the highest scoring areas as simply "in", and all others as "out" (i.e., results are not ranked). It also includes all counties in the State with resources that match the above criteria. Given that wildfire hazard was weighted highest in the CFIP priority landscape composite, a revised analysis using 2020 post-wildfire data is likely to highlight areas significantly different from those shown in the 2007-era results.

Discussion

The risk-and-asset based GIS layers discussed in this document, targeted for the specific needs and criteria of several CAL FIRE programs, present prioritizations statewide based upon well-defined, more localized landscape characteristics rather than regions. Although this diverges from the being based on classically defined "regions", we believe that taken together these priority landscapes meet the intent of the region-based requirement of Forest Action Plan, and in fact provide a more detailed and useful tool to help inform decision-makers in program grant reviews and approval processes. Priority regions, where desired, can easily be derived from a spatial synthesis (summarization) of priority landscape GIS layers.

The priority landscape GIS layer approach also facilitates web-based information consumption, and maintenance and ease of updating of priority areas as conditions on the ground change. The year 2020, with its record-breaking wildfire season, illustrates the advantages of this strategy. As we revise our data inputs where needed, we will re-create new priority landscapes to reflect those changes.

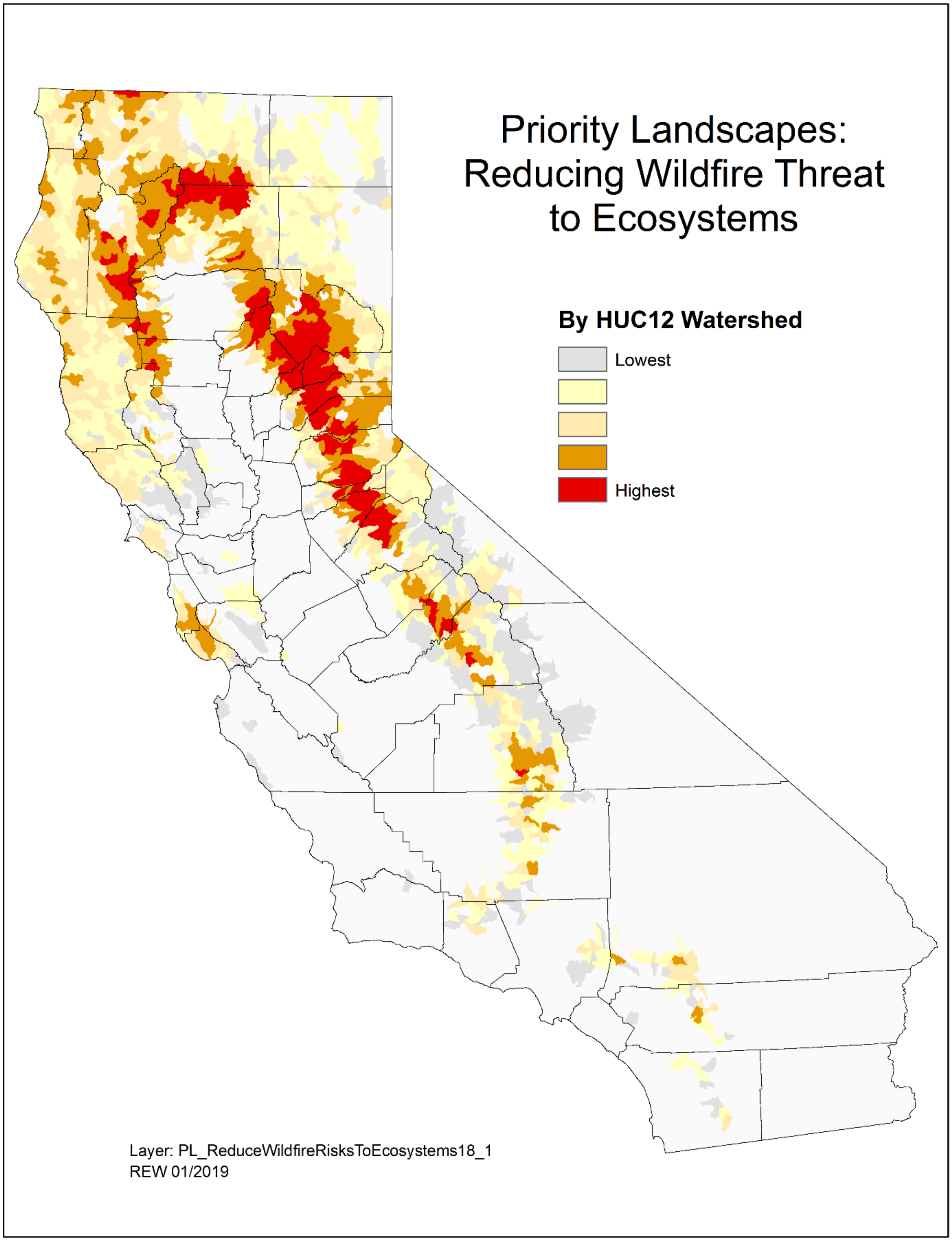


Figure 1. Reducing Wildfire Threats to Forest Ecosystem Services (2018)

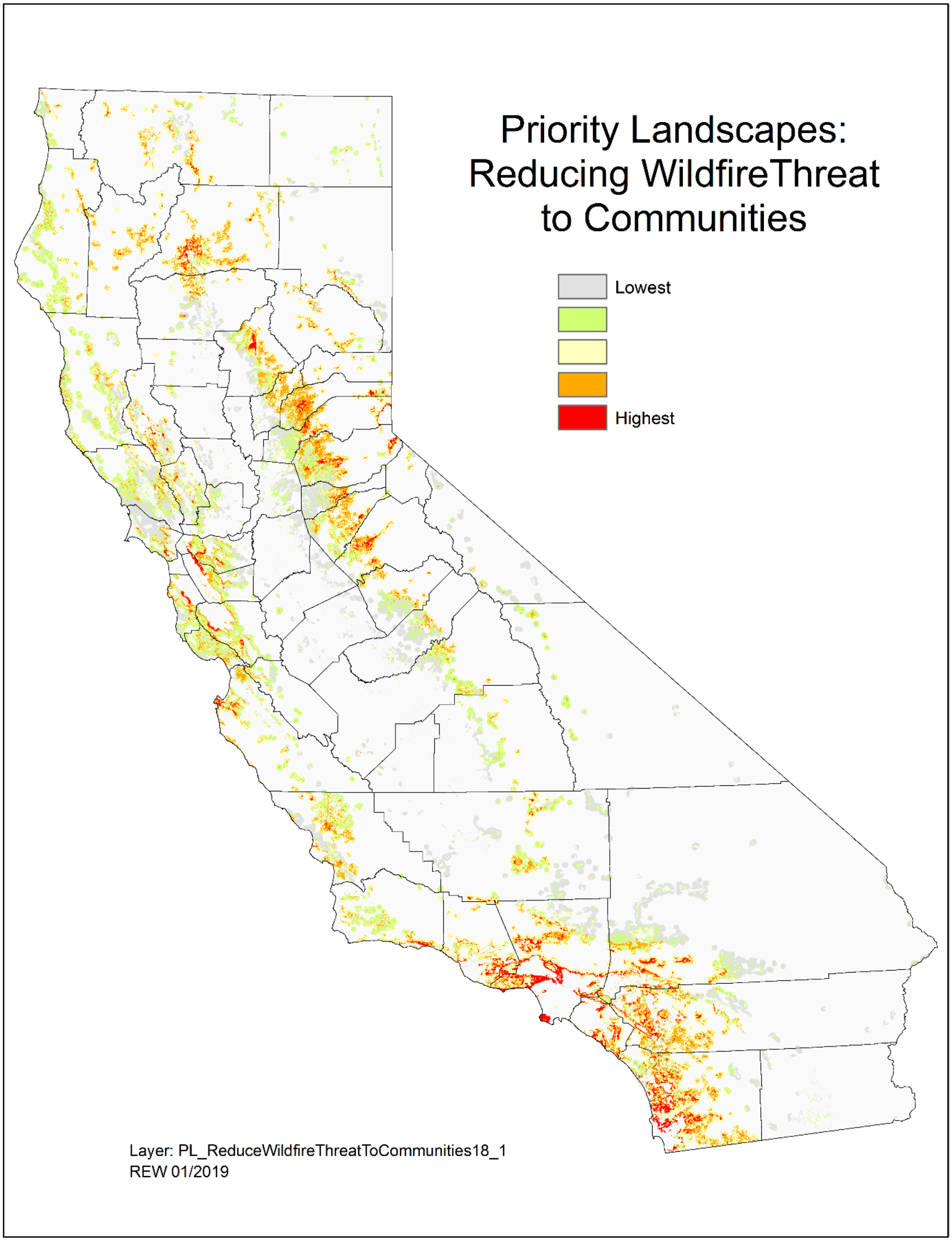


Figure 2. Reducing Wildfire Threat to Communities (2018)

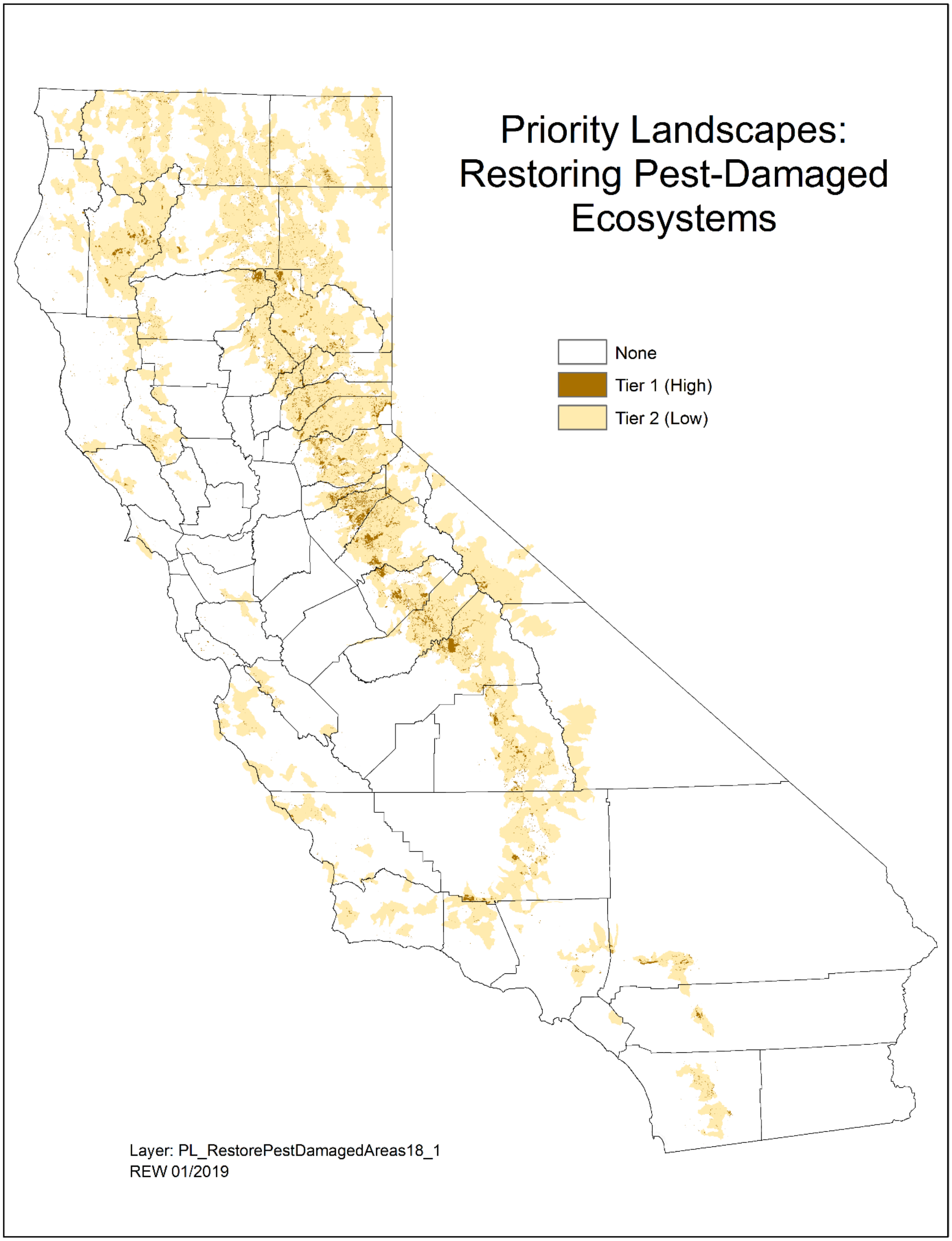


Figure 3. Restoring Pest-Damaged Forested Ecosystems (2018)

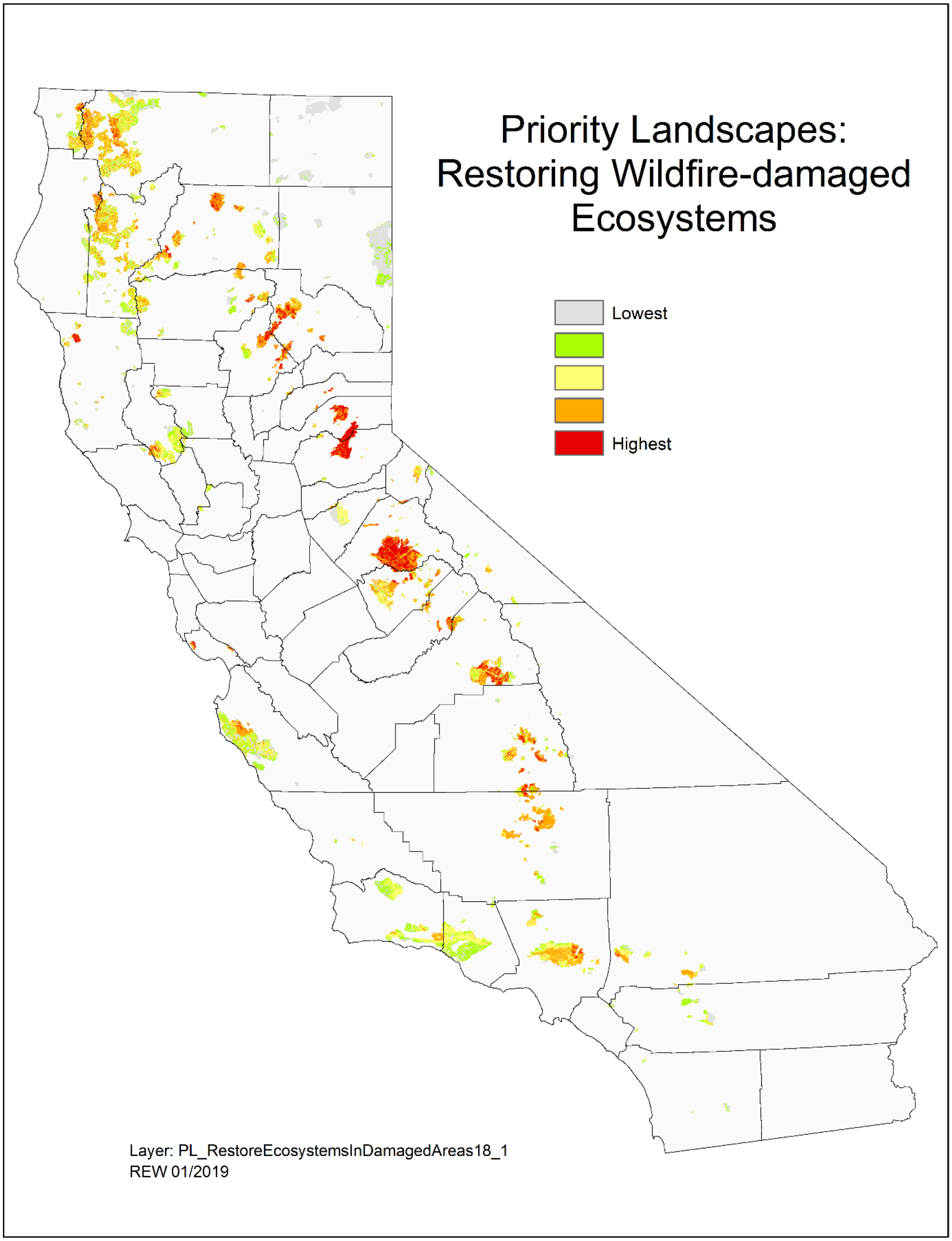


Figure 4. Restoring Wildfire-Damaged Forest Ecosystems (2018)

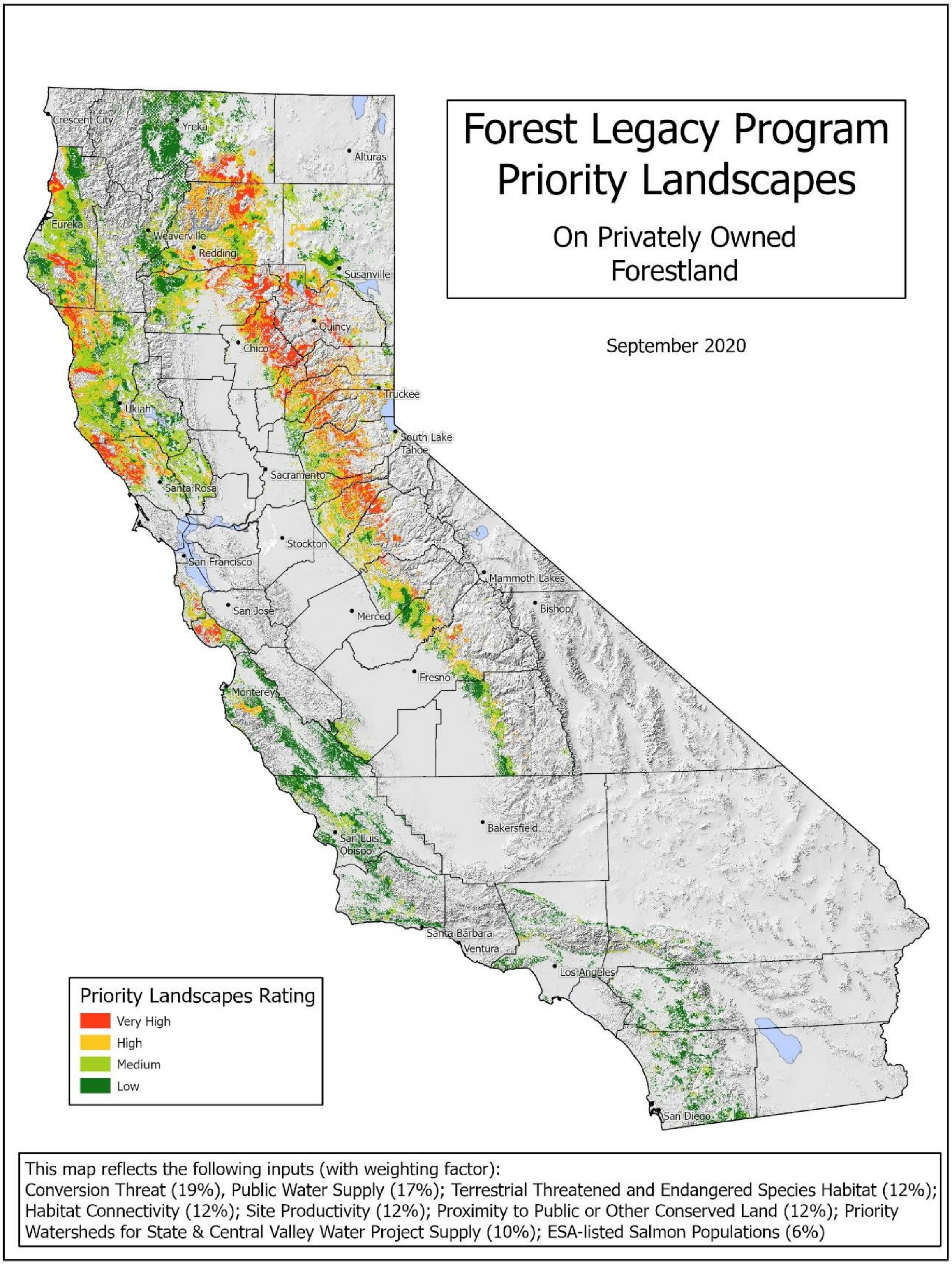


Figure 5. Forest Legacy Priority Landscapes (from Assessment of Need, 2020)

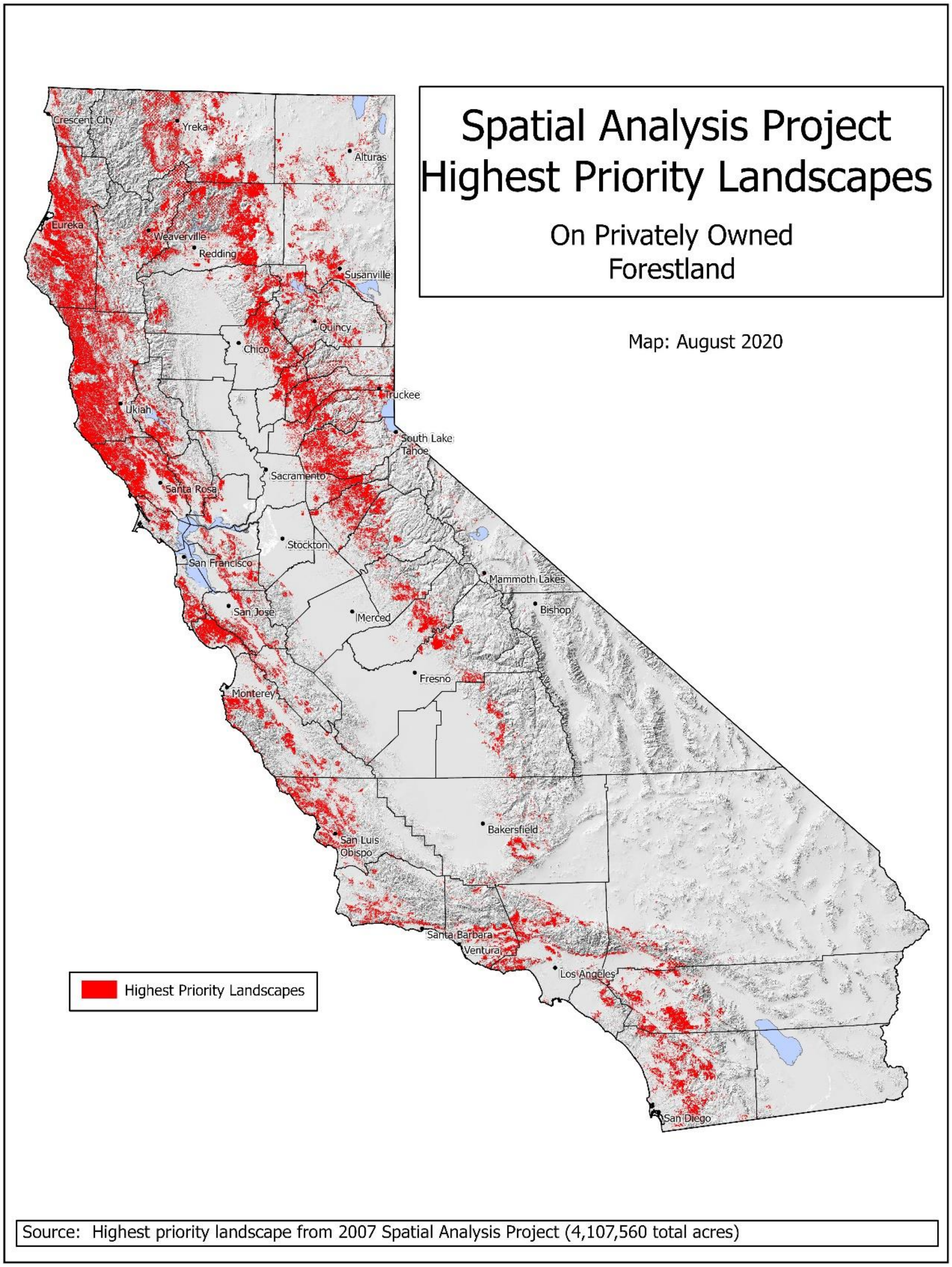


Figure 6. Recently updated Spatial Analysis Project results for California (2020).